



Research paper

Deep learning for distributed acoustic sensing-based microseismic candidate-window classification in enhanced geothermal systems: A comparative study of twelve architectures

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ABSTRACT

Microseismic event discrimination in distributed acoustic sensing data is critical for monitoring enhanced geothermal system operations, yet classical detectors based on short-term to long-term energy ratios, in the simple validation-tuned form evaluated here, fail to discriminate genuine seismic arrivals from structured noise. This paper presents a comparative study of twelve artificial intelligence models based on deep learning, spanning five architecture families—convolutional neural networks, transformers, recurrent networks, state-space models, and graph neural networks—benchmarked against three classical baselines on 3974 manually labeled distributed acoustic sensing recordings from a field stimulation campaign. We contribute an open-source labeling tool for multichannel seismic data, and evaluate complete 1.2-second triggered candidate-window classification rather than end-to-end continuous-stream detection. A preliminary temporal-context ablation is reported but not treated as a repeated-seed conclusion. The squeeze-and-excitation residual network and the plain residual network are statistically comparable; the squeeze-and-excitation variant has the highest numerical F1-score (the harmonic mean of precision and recall), 98.50%, and an area under the receiver operating characteristic curve of 99.46%, while the plain residual network has slightly stronger balanced accuracy and noise recall under the exported reviewer metrics. A selective state-space model achieves 95.76% F1-score, providing a competitive baseline for this architecture family in this dataset. A graph neural network exploiting fiber spatial topology achieves 95.47% F1-score with only 94,290 parameters. Gradient-weighted class activation mapping provides qualitative evidence that the convolutional models focus on primary-wave moveout patterns. To stress-test external transfer, all twelve saved Stage 14 models are evaluated without fine-tuning on Utah Frontier Observatory for Research in Geothermal Energy April 2024 data, a different acquisition. The external set contains 4073 candidate windows (2024 event windows and 2049 non-event windows). The native records have 1496 traces and 4800 samples at 4000 samples per second; the same preprocessing function was used for event and non-event windows: the first 361 traces were retained, the records were downsampled to 2400 samples at 2000 samples per second using anti-aliased polyphase resampling, and per-channel z-score (zero mean, unit variance) normalization was applied. The best external model is the convolutional neural network-gated recurrent unit hybrid with F1-score 0.9930, balanced accuracy 0.9931, and Matthews correlation coefficient 0.9863. We interpret this as zero-shot evidence of matched-pipeline external robustness, not as universal cross-site validation or deployment-ready generalization. A three-seed retraining of the top two residual convolutional models with full random-state control further confirms that the small numerical F1 advantage of SE-ResNet over ResNet is within initialization variance: the pooled inter-seed standard deviation ($\sigma \approx 0.0036$ F1) is approximately four times the inter-model mean difference (0.0009 F1), and paired *t*-tests and Wilcoxon signed-rank tests fail to reject the null of no difference on F1, balanced accuracy, and MCC.

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1. Introduction

Microseismic monitoring is an essential tool in enhanced geothermal system (EGS) operations, providing real-time information about hydraulic fracture propagation, fault activation, and induced seismicity hazard (Rashid et al., 2025; Wu et al., 2025). Distributed acoustic sensing (DAS) has emerged as a high-density, continuous sensing modality for microseismic monitoring, offering spatial coverage and temporal resolution that is impractical with conventional geophone arrays (Ben-Zeev and Lior, 2026). DAS records axial strain or strain rate along a fiber-optic cable deployed in a borehole, producing multichannel waveform data at 1000–10000 Hz with channel spacings of 1–5 m.

Despite this data richness, deployed monitoring systems still rely primarily on the classical short-term average over long-term average (STA/LTA) detector (Shearer, 2009), which computes per-channel energy ratios and cannot exploit the spatial coherence patterns that distinguish genuine seismic arrivals from structured noise. In EGS environments, DAS recordings are contaminated by pump harmonics, electrical interference, and optical loading transients that produce amplitude excursions comparable to genuine microseismic arrivals on individual channels. Under these conditions, STA/LTA routinely achieves false alarm rates approaching 100%, as demonstrated in this study and consistent with findings by Yu et al. (2024) at Utah FORGE, where DASEventNet discovered five times more events than STA/LTA. The Short-Time Energy (STE) and Matched Filter detectors offer marginal improvements but remain fundamentally limited: STE cannot suppress structured periodic noise, and template-based Matched Filters fail when microseismic waveform diversity is high (Mousavi and Beroza, 2022). These limitations of classical detectors motivate the systematic exploration of deep learning architectures that can jointly model spatial coherence and temporal waveform structure across the complete triggered multichannel DAS candidate window.

Applying supervised deep learning to this problem, however, requires reliably labeled training data—and for DAS microseismic detection, this requirement presents a practical bottleneck. DAS data are stored as multichannel SEG-Y files that general-purpose labeling tools (designed for images or tabular data) cannot display with the gain control, normalization, and spatial coherence visualization that domain experts need for confident event/noise discrimination. Existing DAS studies either label data using ad hoc scripts without releasing the tool (Yu et al., 2024; Boitz and Shapiro, 2024) or rely on cross-referencing with geophone catalogs (Huot et al., 2022), which limits reproducibility and restricts labeling to sites where co-located conventional instrumentation exists. Furthermore, publicly available DAS microseismic benchmarks with binary event labels remain scarce: the Utah FORGE catalogs (Dvory et al., 2025; Zhu et al., 2026) provide event times but not pre-segmented, labeled DAS recordings ready for classifier training. These infrastructure gaps—the absence of a reusable, open-source DAS labeling tool and of a labeled field benchmark—constrain progress across the community and motivate the first two contributions of this work.

With these data infrastructure challenges in mind, we turn to the architectural question at the core of this study. Recent deep learning work has demonstrated substantial improvements over classical detectors for DAS microseismic detection: Wamriew et al. (2021) showed that CNNs trained on synthetic DAS data with real noise can simultaneously detect, locate, and invert for velocity models; Boitz and Shapiro (2024) achieved completeness to $M_w - 1.6$ with a U-Net; Yu et al. (2024) presented DASEventNet, achieving 100% accuracy on a balanced test set from Utah FORGE; and the EQTransformer (Mousavi et al., 2020) demonstrated that attention-based architectures can reach state-of-the-art performance on conventional seismometer data. However, each of these studies evaluates a single architecture or a narrow set of architectures against STA/LTA alone, leaving the practitioner without evidence-based guidance on which architectural family is most appropriate for a given triggered screening scenario.

This is not merely an academic gap. DAS microseismic monitoring involves fundamentally different data characteristics from conventional seismometer data: the input is a dense, regularly sampled spatial array (hundreds of channels at meter-scale spacing) rather than a sparse, irregularly distributed station network; the signals exhibit anisotropic frequency–wavenumber structure imposed by gauge-length filtering; and the noise environment is dominated by coherent industrial artifacts rather than ambient seismicity. These properties create distinct inductive bias requirements that may favor certain architecture families over others—for example, CNNs with asymmetric kernels may better capture anisotropic moveout than isotropic attention, while GNNs that encode the known fiber topology (Shahabudin et al., 2024) may achieve competitive performance with far fewer parameters by exploiting the spatial structure *a priori*. A systematic comparison across architectural families is therefore necessary not because no one has performed it, but because the physics of DAS data creates an open question about which inductive biases best match the signal structure in triggered candidate windows—a question whose answer informs triggered candidate-window classifier design for EGS monitoring workflows.

Two architecture families with strong theoretical motivation for DAS data further sharpen this question. Mamba selective state-space models (Gu and Dao, 2023) offer linear-time sequence modeling with input-dependent selectivity well-suited to detecting brief transient arrivals in long background windows, yet have not been applied to DAS microseismics. Graph neural networks have recently been shown to achieve high detection accuracy on DAS microseismic data by Shahabudin et al. (2024), who demonstrated that graph representations of fiber topology capture spatial coherence patterns relevant to event discrimination; however, their study did not benchmark GNN performance against CNN, transformer, RNN, or SSM alternatives on the same dataset, leaving the relative positioning of GNNs within the broader architectural landscape unresolved.

Building on these observations, this paper addresses the data infrastructure, architectural comparison, and untested-family gaps with five contributions:

- (1) The *Advanced Gather Viewer*: an open-source, purpose-built SEG-Y labeling tool for multichannel DAS data that provides the gain control, spatial coherence visualization, and structured export workflow required for reliable manual labeling of triggered DAS recordings (see public repository information note below).
- (2) A manually reviewed binary-labeled DAS microseismic benchmark derived from the Cape enhanced geothermal system Frisco 2-P dataset, with consensus labels from two independent domain experts.
- (3) A systematic comparative study of twelve deep learning models across five architectural families (CNN, Transformer, RNN, SSM, GNN), benchmarked against three classical baselines (STA/LTA, STE, Matched Filter), with full statistical significance testing (McNemar + DeLong + Bonferroni over 78 pairs), fixed-checkpoint split-stability and approximate order-sensitivity audits, and a matched-pipeline zero-shot external evaluation on Utah FORGE April 2024 data, providing evidence-based guidance for triggered candidate-window classifier design under this dataset and protocol.
- (4) A Mamba selective state-space model baseline for DAS microseismic detection ($F1 = 95.76\%$), providing an SSM benchmark in this dataset.
- (5) DAS-GNN: a graph-based detector that extends prior GNN work on DAS data (Shahabudin et al., 2024) by introducing explicit k -nearest-neighbor fiber topology encoding with learned spatial coherence propagation via GCN layers, achieving $F1 = 95.47\%$ with only 94,290 parameters—the most parameter-efficient competitive architecture in the study (38× fewer parameters than SE-ResNet)—and providing a direct comparison of GNN against CNN, transformer, RNN, and SSM families on the same DAS benchmark.

Note: The source code for the Advanced Gather Viewer is publicly available at <https://github.com/shokor103072/advanced-gather-viewer>.

2. Related work

2.1. DAS-based microseismic detection

Deep learning for DAS microseismic detection has progressed rapidly in recent years. Wamriew et al. (2021) demonstrated the feasibility of using deep neural networks for simultaneous event detection, location, and velocity model inversion from DAS data acquired during hydraulic fracturing at the Utah FORGE site, training on synthetic data contaminated with real ambient noise. Huot et al. (2022) applied a CNN with continuous wavelet transforms to DAS fiber data, successfully identifying low-amplitude events overlooked in manual analysis. Boitz and Shapiro (2024) developed a U-Net CNN for microseismic event detection in DAS data from the 2019 Utah FORGE stimulation, incorporating geophone information for training and achieving a completeness magnitude of $M_w \sim -1.6$. Yu et al. (2024) presented DASEventNet for Utah FORGE DAS detection, achieving 100% accuracy on a balanced test set and discovering over 5700 new microearthquakes from continuous DAS recordings-five times more than STA/LTA—with a detection threshold of $M_w - 1.80$. Ma et al. (2023) proposed a machine-learning-assisted processing workflow for multi-fiber DAS microseismic data, addressing the challenges of processing multi-well fiber-optic deployments. Al-Hemyari et al. (2024) used a ResNet50 architecture trained on synthetic data paired with real noise recordings for supervised detection and location of passive seismic events recorded with DAS. Mousavi and Beroza (2022) provided a comprehensive review of deep learning in seismology, noting that DAS data volumes (approximately 650 GB/day for a 2 km cable at 2000 Hz) necessitate automated processing methods. Related work also addressed DAS denoising (Lapins et al., 2024; T. Huang et al., 2024), noise removal (Gu et al., 2024), and strain-to-velocity conversion (Al-Qadasi et al., 2026). DAS measurement physics introduces gauge-length filtering, orientation sensitivity, and coupling heterogeneity (Wu et al., 2025; Rashid et al., 2025) that affect amplitude, bandwidth, and noise structure.

Despite this progress, existing studies typically evaluate a single architecture against STA/LTA or a small number of classical baselines, and no comprehensive multi-family comparison-covering CNNs, transformers, RNNs, state-space models, and graph neural networks—has been conducted for DAS microseismic data.

2.2. Attention and channel weighting

Squeeze-and-excitation (SE) networks (Hu et al., 2018) learn channel-wise recalibration weights through a global-average-pooling squeeze operation followed by a two-layer MLP excitation pathway. SE blocks have been integrated into ResNet (He et al., 2016), Inception, and MobileNet backbones with consistent improvements across image classification benchmarks at minimal computational overhead. In seismology, attention mechanisms have been adopted for multiple tasks. Ku et al. (2021) applied attention-based CNNs for earthquake event classification. Jin et al. (2024) combined CNNs with attention mechanisms for microseismic event recognition and transfer learning. Y. Huang et al. (2024) proposed a 1D CNN with efficient channel attention and light inception blocks for classifying seismic events, achieving 96.79% accuracy. In seismic signal enhancement, Saad et al. (2021) used deep learning to recover microseismic signals from noisy data for high-fidelity source-location imaging, and Cui et al. (2024) developed ground-truth-free 3D seismic denoising using channel attention mechanisms. For DAS, where fiber coupling quality varies along the wellbore, SE attention can adaptively suppress poorly coupled channels—a physically motivated application that has not been explored for DAS microseismic classification. The Conformer (Gulati et al., 2020),

originally developed for speech recognition, interleaves local CNN patterns with global multi-head self-attention, capturing both local and long-range dependencies. The EQTransformer (Mousavi et al., 2020) demonstrated that transformer-based architectures can achieve state-of-the-art performance on earthquake detection and phase picking from conventional seismometer data. Neither SE attention nor Conformer has been systematically compared against SSM and GNN baselines on DAS microseismic data.

2.3. State-space models

The S4 model (Gu et al., 2022) introduced structured state-space sequence modeling with $O(L \log L)$ complexity, achieving strong performance on the Long Range Arena benchmark. Mamba (Gu and Dao, 2023) extended this with input-dependent (selective) parameters that allow the model to focus on informative time steps while ignoring irrelevant background, achieving competitive performance on language and audio tasks with linear $O(L)$ complexity. Mamba-2 (Dao and Gu, 2024) further refined the framework by suggesting a theoretical connection between selective state-space models and attention, demonstrating that SSMs can be viewed as structured variants of linear attention. In remote sensing, Mamba-based architectures have attracted rapid interest for their ability to process high-resolution spatial sequences with linear complexity (Chen et al., 2024), and a recent survey (Li et al., 2026) synthesized evidence on when Mamba improves accuracy versus when CNN or transformer designs remain sufficient. In geophysical applications, Diff-Mamba (Zhang et al., 2026) has been applied to well-log reconstruction, demonstrating the utility of selective state-space modeling for geoscientific sequential data. However, no application of Mamba or any SSM architecture to DAS microseismic detection has been published.

2.4. Graph neural networks for seismic data

GNNs model data structured as graphs and have been increasingly applied in seismology for tasks where spatial relationships among sensors are physically meaningful. McBrearty and Beroza (2023) developed a Graph Neural Network associator (GENIE) for earthquake phase association that simultaneously predicts source localization and source-arrival association likelihoods across arbitrary station geometries, re-detecting 96% of USGS events $M > 1$ and discovering 4.2× more events than the standard catalog. Si et al. (2024) proposed PLAN, an all-in-one GNN framework that jointly performs phase picking, association, and location while enforcing cross-station and cross-task physical consistency, achieving improved performance over separate deep learning approaches. Zhang et al. (2022) developed a spatiotemporal GNN (STGNN) that leverages both geographic and waveform information from multiple stations for earthquake localization and magnitude estimation. Hourcade et al. (2025) applied GNNs to prompt elastogravity signals for rapid earthquake characterization, showing advantages of graph-based approaches over CNN-based methods for multi-station seismic data. However, all existing seismological GNN applications model station networks where nodes represent geographically distributed sensors. DAS fiber presents a fundamentally different graph structure: a linear array of densely spaced channels (~2 m) along a single cable, where the spatial topology is known *a priori* and edges connect physically adjacent fiber segments. Shahabudin et al. (2024) bridged this gap by applying GNNs directly to DAS microseismic data, demonstrating that graph representations of fiber topology achieve high detection accuracy by capturing the spatial coherence patterns that distinguish genuine seismic arrivals from noise. Their work established that GNNs are a viable and effective architecture family for DAS microseismic classification. The present study builds on this foundation in two specific ways: first, DAS-GNN introduces explicit k -nearest-neighbor fiber topology encoding with learned spatial coherence propagation via GCN layers (Kipf and Welling, 2017),

Table 1

Comparison of representative DAS microseismic detection studies. Arch. = architecture family evaluated; #Models = number of deep learning models compared; Labels = label/annotation source; Size = number of labeled DAS recordings used; XStage = cross-stage or cross-site evaluation reported.

Study	DAS Site	Arch.	#Models	Labels	Size	XStage
Wamriew et al. (2021)	Utah FORGE	CNN	1	Syn.+noise	70k	No
Huot et al. (2022)	Shale HF	CNN	1	Manual	13,572	No
Boitz and Shapiro (2024)	Utah FORGE	CNN (U-Net)	1	Geophone	~150 evt.	No
Ma et al. (2023)	Multi-fiber	CNN, U-Net	2	STA/LTA+manual	3,584+	No
Al-Hemyari et al. (2024)	CO2CRC	CNN (ResNet50)	1	Syn.+noise	12,364+	No
Yu et al. (2024)	Utah FORGE	CNN (ResNet50)	1	Catalog+noise	2584	No
This work	Cape EGS; Frisco 2-P; FORGE April 2024	CNN, Trans., RNN, SSM, GNN	12	2-expert manual + catalog	3974 + 4073	Yes

achieving competitive performance with only 94,290 parameters (38× fewer than the best CNN); and second, by embedding DAS-GNN within a comprehensive five-family comparison, we provide a direct assessment of where GNNs rank relative to CNN, transformer, RNN, and SSM alternatives on the same benchmark—answering the open question of whether the parameter efficiency advantage of graph-based DAS detection comes at the cost of detection accuracy.

2.5. Summary and positioning

Table 1 summarizes the key characteristics of representative prior studies and positions the present work. Existing DAS microseismic detection studies evaluate one or two architectures; none benchmarks across all five families or includes SSM or GNN baselines for DAS data.

Beyond architecture coverage, prior DAS and seismic detection studies differ substantially in how they evaluate detectors, which complicates direct comparison. Reported pipelines vary in whether they operate on continuous streams or on pre-triggered candidate windows; in how labels are obtained (manual annotation, geophone-catalog cross-referencing, or synthetic events with added noise) and how label uncertainty is treated; in whether decision thresholds are tuned on validation or test data; in whether class imbalance is reported with imbalance-aware metrics; in whether uncertainty is quantified at all; and in whether false alarms are reported as confusion counts, rates per window, or operational triggers per unit time. Cross-site or cross-installation validation is rarely reported. The present study is explicit about each of these choices: it addresses pre-triggered 1.2-second candidate windows; uses two-expert consensus manual labels with disagreement statistics; tunes all thresholds on validation data only; reports balanced accuracy, Matthews correlation coefficient, per-class recall, and confusion counts alongside bootstrap confidence intervals and calibration; and confines false-alarm interpretation to the triggered-candidate setting rather than converting it to an operational trigger rate. These protocol choices, rather than headline accuracy alone, determine whether a reported comparison is informative, and we state them explicitly so that the present results can be compared on equal terms with future work.

3. Methodology

Fig. 1 presents an overview of the experimental pipeline, which proceeds through five stages: (1) data acquisition and labeling using the Advanced Gather Viewer; (2) preprocessing, quality filtering, and stratified splitting; (3) model training across twelve deep learning architectures and three classical baselines; (4) evaluation using imbalance-aware metrics, confusion matrices, calibration, and threshold sensitivity; and (5) validation through statistical significance testing, uncertainty analysis, fixed-checkpoint split-stability audits, zero-shot external cross-site evaluation, and Grad-CAM physical interpretation. Each stage is detailed in the subsections below.

3.1. Data source

The dataset is the Cape EGS Frisco 2-P Well Stimulation Microseismic Data (Dadi and Titov, 2024), collected during stimulation of the Frisco 2-P well by Fervo Energy from June 1–11, 2024, near the Utah FORGE geothermal site, Milford, Utah. Microseismic data were collected using distributed acoustic sensing fiber in well 16B and three three-component geophones located in wells 56–32, 78B, and 32. The available dataset metadata report a 2000 Hz temporal sampling rate, approximately 2 m distributed acoustic sensing channel spacing, and 1.2-second record duration. Gauge length and interrogator model are not reported in the available metadata. Each SEG-Y file is a triggered candidate recording; the first nine traces are described in the dataset metadata as geophone traces and the remaining traces as distributed acoustic sensing traces. Each triggered SEG-Y file contains 361 traces in total (nine geophone traces followed by 352 distributed acoustic sensing traces); all 361 traces were retained directly, so no signal-to-noise-favorable channel subset was selected from a larger array, and the model input is predominantly (352 of 361 channels) distributed acoustic sensing data with the nine leading geophone channels included. We refer to this dataset as “Stage 14” throughout the paper (a specific stimulation-stage identifier from the Frisco 2-P campaign, not a numeric range), and use it for all in-domain training, validation, and testing.

This study evaluates classification of complete 1.2-second triggered candidate windows and does not evaluate end-to-end continuous-stream detection. After quality control, Stage 14 contains 3974 samples: 3112 event windows and 862 noise windows. To stress-test generalization beyond this acquisition, we additionally evaluate the trained models *zero-shot* on an external dataset—Utah FORGE April 2024 data (Utah FORGE, 2024). The external set contains 4073 folder-derived candidate windows: 2024 event windows and 2049 non-event windows. Its preparation and use are described in Section 7.9. This analysis is a zero-shot external evaluation that supports matched-pipeline external robustness, not universal cross-site validation or deployment-ready generalization. Gauge length and interrogator model remain unreported for the Stage 14 training data.

3.2. Labeling tool

Manual labeling used an open-source SEG-Y labeling tool (Rahaman, 2024) (MIT license, Go + web frontend). The tool provides multichannel gather display with adjustable gain, trace normalization, and color mapping; keyboard-driven binary event/non-event labeling with automatic file advance; structured CSV export with file name, timestamp, file size, and reviewer metadata; and session resumption with progress tracking. All labeled files were reviewed independently by two domain experts. Each file was examined as a multichannel DAS gather and assigned a binary event/non-event label after visual inspection of waveform morphology, spatial coherence across channels, and signal-to-noise characteristics. Stage 14 (4189 files) was completed in under 6 h.

Inter-expert label agreement. The two experts labeled the dataset independently; disagreements were identified by comparing the two

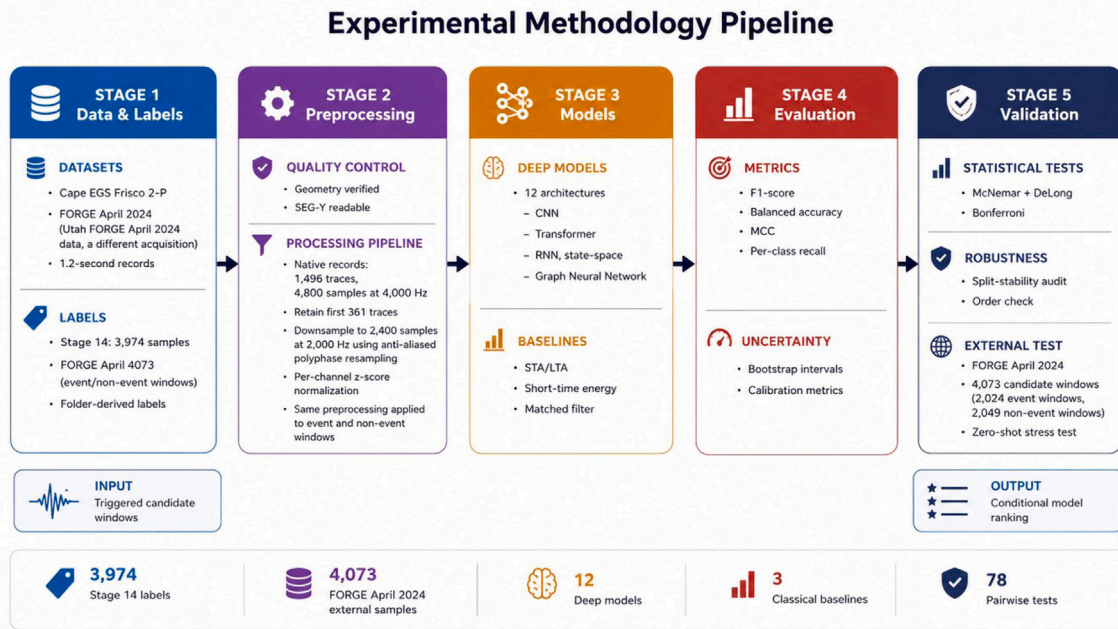


Fig. 1. Overview of the experimental methodology. The pipeline spans data acquisition and labeling (Stage 1), preprocessing and splitting (Stage 2), model training across five architectural families and three classical baselines (Stage 3), performance evaluation (Stage 4), and multi-protocol validation including statistical testing, uncertainty analysis, fixed-checkpoint split-stability audits, exploratory zero-shot external cross-site evaluation, and Grad-CAM interpretation (Stage 5).

CSV exports file-by-file before consensus review. The two reviewers agreed on 4141 of 4189 files (raw agreement rate: 98.9%; 48 disputed files, 1.1%). Disputed files were re-examined jointly and assigned a consensus label. Disagreements were confined to very-low-SNR borderline arrivals (peak-to-peak amplitude $< 2\sigma$ above background noise floor), consistent with the completeness boundary of STA/LTA-triggered catalogs. The pre-consensus raw agreement rate of $> 98.9\%$ and the restriction of disagreements to a physically well-defined ambiguity regime indicate that the benchmark labels are reliable.

Non-event taxonomy. Non-event files fell into three recurring categories: (i) pump harmonic noise (periodic, spatially coherent amplitude modulation at 4–16 Hz across all channels, no apparent moveout); (ii) optical loading transients (sudden amplitude step change across all channels simultaneously, no temporal dispersion); and (iii) ambient noise triggers (stationary, no coherent arrival, triggered by brief amplitude excursions from industrial surface activity). These three categories constitute 97% of all non-event labels.

3.3. DAS sensing physics and remote sensing context

DAS functions as a distributed remote sensor: it continuously measures axial strain or strain rate along a fiber-optic cable by detecting Rayleigh backscatter phase shifts induced by seismic wave passage. This positions DAS as a fiber-optic remote sensing modality (Rashid et al., 2025), distinct from conventional single-point geophones in three physically important ways.

Gauge-length filtering. A distributed acoustic sensing measurement integrates strain over a finite gauge length G , acting as a spatial low-pass filter with a notch at apparent wavelengths equal to G . The gauge length for this dataset is not reported in the available metadata, so we do not model the instrument response quantitatively. More generally, the frequency–wavenumber content of distributed acoustic sensing microseismic arrivals is anisotropic, which motivates testing CNN stems with asymmetric spatial–temporal kernels.

Apparent velocity and moveout. A seismic P-wave arriving at the fiber at incidence angle θ produces an apparent velocity $v_{app} = v_P / \sin \theta$

along the fiber axis. For typical EGS microseismic events at source-receiver offsets of 50–500 m and P-wave velocities of 4–6 km/s, apparent velocities of 3–15 km/s produce moveout delays of 0.1–3 ms per channel spacing. The Grad-CAM attention patterns in Fig. 3 are therefore interpreted visually as moveout-like structure, not as quantitative arrival picks or apparent-velocity estimates.

Fiber coupling heterogeneity. The sensitivity of a cemented borehole DAS fiber varies with local formation coupling quality, cement bond, and fiber orientation relative to the incident wavefield. This produces channel-dependent SNR variations that are quasi-stationary over the duration of a stimulation stage. The SE channel attention mechanism in SE-ResNet can suppress low-SNR feature channels and amplify informative feature channels — a physically plausible operation that plain ResNet may perform only implicitly through its learned weights. Because SE-ResNet and ResNet are statistically and practically comparable under the exported tests, the small numerical F1 difference should be interpreted as consistent with, but not proof of, a channel-recalibration benefit.

3.4. Labels, preprocessing, and splits

Of 4189 labeled files, 3112 (74.3%) were labeled *event* and 1077 (25.7%) *non-event*, giving a raw event-to-noise ratio of 2.89:1. After removing corrupted and incomplete SEG-Y files (215 non-event files failed channel-count or sample-count validation), the final dataset contained 3112 events and 862 noise recordings (3974 total), with a cleaned event-to-noise ratio of 3.61:1 (referred to hereafter as 3.6:1). Preprocessing: (i) the first 361 traces were retained directly from each Stage 14 SEG-Y file (these comprise all traces present in each file: the nine leading geophone traces and the 352 distributed acoustic sensing traces, so no favorable channel subset is selected); (ii) records were padded or trimmed to 2400 time samples; (iii) per-channel z-score normalization was applied (zero mean, unit variance); (iv) the final Stage 14 array had shape (3974, 1, 361, 2400). Per-channel z-score normalization stabilizes training but suppresses absolute amplitude differences, so the results should not be interpreted as using calibrated absolute amplitudes. Split: 70%/15%/15% (stratified, seed 42) yielding

Table 2

Recovered training settings for the twelve deep models. Settings are reconstructed from code because no standalone hyperparameter search log was found. Source abbreviations: comp., cnn, five, and vit denote the corresponding training scripts.

Model	Params	Opt.	LR	Batch	Epochs	Stop	Sched.	Loss/source
SE-ResNet	3.5846M	AdamW	3e-4	16	50	Val. F1; p12	Cos. 50	CE/comp.
ResNet	3.5618M	AdamW	3e-4	16	60	Val. F1; p12	Cos. 60	CE/cnn
Conformer	9.8751M	AdamW	3e-4	8	50	Val. F1; p12	Cos. 50	CE/comp.
MBHCT	2.0870M	AdamW	3e-4	16	50	Val. F1; p12	Cos. 50	CE/comp.
Mamba-DAS	0.8586M	AdamW	3e-4	8	50	Val. F1; p12	Cos. 50	CE/five
CNN-GRU	2.1085M	AdamW	3e-4	8	50	Val. F1; p12	Cos. 50	CE/five
DAS-GNN	0.0943M	AdamW	3e-4	8	50	Val. F1; p12	Cos. 50	CE/five
ConvNeXt	1.9027M	AdamW	3e-4	16	50	Val. F1; p12	Cos. 50	CE/comp.
CNN-BiLSTM	2.3551M	AdamW	3e-4	8	50	Val. F1; p12	Cos. 50	CE/comp.
S4-SSM	0.2961M	AdamW	3e-4	16	50	Val. F1; p12	Cos. 50	CE/five
ViT	18.5251M	AdamW	1e-4	8	60	Val. F1; p15	Warm+cos.	CE+LS+mixup/vit
GRU	2.0760M	AdamW	3e-4	8	50	Val. F1; p12	Cos. 50	CE/five

2781/596/597 train/validation/test samples. Source-file provenance was reconstructed from the dataset-preparation script order because the saved arrays do not embed an original manifest. The 3.6:1 class imbalance was addressed by weighted random sampling.

3.5. Training protocol

All deep learning models receive inputs of shape (1,361,2400). Hyperparameters were reconstructed from the training scripts. Most models used the same stratified 70/15/15 split, AdamW optimizer, weighted random sampling, validation-F1 checkpoint selection, cosine annealing schedule, and early stopping. No standalone hyperparameter-search log or equal-budget tuning ledger was available. The comparison is therefore best interpreted as a controlled common-protocol benchmark under the available code-level settings, not as an exhaustive equal-budget architecture search. The recovered per-model training settings are summarized in Table 2.

3.6. Family A — Convolutional neural networks

SE-ResNet (highest numerical F1): ResNet backbone (He et al., 2016) augmented with SE channel attention (Hu et al., 2018) after each residual stage. SE compresses spatial information via global average pooling, produces per-channel recalibration weights via a two-layer MLP bottleneck (ratio 16), and rescales feature maps channel-wise. This mechanism is physically motivated for DAS data: fiber coupling quality varies along the wellbore, producing channel-dependent SNR variations, and SE attention can adaptively suppress poorly coupled channels while amplifying informative ones. Stem: 7×15 kernel, stride (2,4); four SE-residual stages (32/64/128/256 channels); MLP head with 50% dropout. The asymmetric 7×15 stem kernel is designed for DAS data geometry: the spatial axis (7 channels = 14 m) captures local fiber coherence, while the extended temporal axis (15 samples = 7.5 ms at 2000 Hz) spans the expected P-wave arrival duration at sub-millisecond resolution. Parameters: 3.58M.

ResNet: identical backbone without SE attention. Serves as the controlled ablation partner for SE-ResNet, isolating the contribution of channel attention from the residual learning framework. Batch normalization after each convolutional layer; skip connections enable gradient flow through the four residual stages. Parameters: 3.56M.

MBHCT (Rahaman et al., 2025): a multi-branch hybrid architecture with three parallel CNN branches extracting features at different spatial-temporal scales: a temporal branch (1×15 kernel capturing waveform morphology), a spatial branch (15×1 kernel capturing cross-channel coherence), and a joint branch (7×15 kernel capturing coupled spatiotemporal features). Each branch is adaptively pooled to a 4×16 feature map. The branch outputs are flattened into 64 tokens, concatenated along the feature dimension, projected to the transformer embedding size, prepended with a learned class token, and combined

with a learned positional embedding before refinement by a 4-layer transformer encoder (6 heads, embed dim 192). Parameters: 2.09M.

ConvNeXt (Liu et al., 2022): a modernized pure-CNN architecture incorporating design elements from transformers: 7×7 depthwise separable convolutions for large receptive fields, LayerNorm replacing BatchNorm, inverted bottleneck blocks (expand ratio 4 $\times$), GELU activation, and learnable layer scale parameters. This architecture tests whether transformer-inspired CNN design improves DAS detection without attention mechanisms. Parameters: 1.90M.

3.7. Family B — Transformers

Conformer (Gulati et al., 2020): originally developed for speech recognition, the Conformer interleaves local convolution with global self-attention to capture both fine-grained local patterns and long-range dependencies. A CNN front-end downsamples the DAS input to 75 temporal tokens (dim 256), followed by 6 Conformer blocks, each containing a macaron-style feed-forward-MHSA-depthwise-conv-feed-forward sandwich structure. Multi-head self-attention (4 heads) models global temporal relationships, while depthwise convolution (31×1 kernel) captures local patterns within the attention context. This dual local-global design is well-suited for DAS data where events exhibit both local amplitude peaks and global moveout patterns. Parameters: 9.88M.

ViT (Dosovitskiy et al., 2021): the Vision Transformer divides the DAS input into (19,50) non-overlapping patches, yielding 912 tokens. Each patch is linearly projected to 384-dimensional embeddings with learnable positional encoding. A 10-layer transformer encoder (8 heads per layer) processes the token sequence using multi-head self-attention and feed-forward networks. ViT has no convolutional inductive bias, relying entirely on data-driven attention to discover spatial and temporal structure. This makes it a useful baseline for assessing whether attention alone, without spatial priors, can learn DAS wave physics at the available dataset scale. Parameters: 18.53M.

3.8. Family C — Recurrent neural networks

CNN-GRU: a hybrid architecture combining multi-scale spatial feature extraction with temporal sequence modeling. Three parallel CNN branches with kernels of 5×1 , 9×1 , and 15×1 extract spatial features at three scales (10 m, 18 m, 30 m fiber extents), capturing both local channel correlations and broader moveout patterns. The concatenated multi-scale features are fed to a two-layer bidirectional GRU (hidden 256) that models temporal dependencies in both forward and reverse directions. An attention pooling layer weights the GRU hidden states to produce the final classification vector. Parameters: 2.11M.

CNN-BiLSTM: single-scale spatial CNN followed by a two-layer bidirectional LSTM (hidden 256) with attention pooling. Unlike CNN-GRU, the spatial CNN uses a single kernel scale, and the LSTM cell

includes an additional cell state providing longer-term memory capacity through explicit forget gates. This architecture tests whether LSTM's gated memory offers advantages over GRU's simpler update mechanism for DAS temporal modeling. Parameters: 2.36M.

GRU: a pure recurrent baseline without convolutional preprocessing. Per-channel temporal downsampling (4×, 2×) reduces the 2400-sample input to a manageable sequence length, followed by a bidirectional GRU with attention pooling. This architecture serves as a controlled comparison: the performance gap between CNN-GRU and pure GRU isolates the contribution of spatial CNN preprocessing before temporal modeling. Parameters: 2.08M.

3.9. Family D — State-space models

S4-SSM: CNN front-end (150 temporal tokens, dim 128) followed by 4 S4-inspired layers using FFT-based $O(L \log L)$ convolution with diagonal state matrix. Parameters: 296K.

Mamba-DAS[†]: CNN front-end (75 tokens, dim 128) followed by 6 custom Mamba-style selective SSM blocks (Gu and Dao, 2023). Each block uses input-dependent selective parameters, depthwise causal convolution, and SiLU gating. The implementation in this study is a simplified local implementation rather than the official mamba-ssm package with optimized CUDA selective-scan kernels; runtime results for this model should therefore be interpreted as implementation-specific. Parameters: 859K.

3.10. Family E — Graph neural networks

DAS-GNN[†]: models the 361 DAS channels as a spatial graph $G = (V, E)$ where each channel is a node ($|V| = 361$) and edges connect the $k=5$ nearest fiber neighbors at 2 m spacing, encoding the known linear geometry of the borehole fiber as a structural prior. This graph construction is physically motivated: P-wave arrivals produce spatially coherent amplitude patterns that propagate across adjacent channels with characteristic moveout, and the graph topology enables the network to aggregate this coherence information explicitly.

The architecture proceeds in three stages. First, a shared-weight per-channel temporal CNN (three 1D convolutional layers with batch normalization, ReLU, and max pooling) independently extracts a 128-dimensional feature vector from each channel's 2400-sample time series, producing a node feature matrix $\mathbf{H} \in \mathbb{R}^{361 \times 128}$. Second, two graph convolution layers (GCN (Kipf and Welling, 2017)) propagate spatial coherence information along the fiber topology: $\mathbf{H}' = \sigma(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} \mathbf{H})$, where $\hat{A} = A + I$ is the adjacency matrix with self-loops and $\hat{D}$ is the corresponding degree matrix. Each GCN layer aggregates features from the k -nearest fiber neighbors, implementing a learned spatial coherence filter over adjacent channels. Relative timing is preserved implicitly through the per-channel temporal features, but the model is not an explicit physics-constrained moveout model. Third, a global mean readout pools the node features into a single graph-level representation, followed by an MLP classifier (256 → 128 → 2) with 50% dropout.

DAS-GNN achieves this with only 94,290 parameters—38× fewer than SE-ResNet—making it the most parameter-efficient competitive architecture in the study and particularly suitable for resource-constrained triggered screening scenarios where model size and inference cost are constrained. This architecture builds on recent work demonstrating the effectiveness of GNNs for DAS microseismic detection (Shahabudin et al., 2024), extending it with explicit k -nearest-neighbor topology encoding and positioning it within a multi-family comparison framework.

3.11. STA/LTA baseline

The STA/LTA detector (Shearer, 2009) is the most widely deployed classical method in operational microseismic monitoring. It computes a ratio of short-term energy to long-term energy on each channel independently, using a sliding window with STA=50 ms (100 samples) and LTA=500 ms (1000 samples). The per-recording detection score is the maximum STA/LTA ratio across the sampled channel set: $S = \max_c \max_t r_c(t)$, where $r_c(t)$ is the STA/LTA ratio for channel c at time t . The detection threshold τ is optimized on the validation set to maximize F1-score; the resulting threshold is then applied to the test set without further adjustment. This strict validation-only tuning protocol avoids test-set threshold selection. The fundamental limitation of STA/LTA is that it operates on individual channels without cross-channel spatial analysis and therefore cannot distinguish spatially coherent seismic arrivals from structured noise sources that produce comparable amplitude excursions on single channels.

3.12. Short-time energy (STE) detector

The STE detector addresses a limitation of STA/LTA by normalizing short-term energy against recording-level RMS energy, producing a dimensionless score that is less sensitive to absolute amplitude variations across recordings. For channel c with signal $x_c(t)$, the STE score is:

$$s_c = \frac{\sqrt{\max_t \left(\frac{1}{W} \sum_{k=0}^{W-1} x_c(t+k)^2 \right)}}{\sqrt{\frac{1}{T} \sum_t x_c(t)^2}} \quad (1)$$

where $W=200$ samples (100 ms at 2000 Hz) is the short-time window and $T=2400$ is the complete candidate-window length. The numerator captures peak local energy, while the denominator provides recording-level normalization. The recording score is $\max_c s_c$ computed over channels sampled every 36th (11 channels total: 0 through 360) for computational efficiency. A threshold θ is optimized on the validation set to maximize F1-score and applied unchanged to the test set. Like STA/LTA, STE operates on individual channels without spatial coherence analysis, but its energy normalization partially mitigates the amplitude calibration problem that causes STA/LTA to trigger on all high-energy recordings regardless of waveform structure.

3.13. Matched filter detector

The Matched Filter represents the strongest classical baseline evaluated here, incorporating waveform morphology through template cross-correlation. It constructs $N_T=20$ template waveforms from the highest-SNR training events (ranked by per-channel standard deviation), each averaged across 11 sampled channels and normalized to unit ℓ_2 norm. The recording score is the maximum normalized cross-correlation magnitude over all templates and channels:

$$S = \max_{c,k} \max_{\tau} |\langle \hat{x}_c(\cdot - \tau), h_k \rangle| \quad (2)$$

where $\hat{x}_c$ is the ℓ_2 -normalized recording trace, h_k is template k , and τ is the time lag. The detection threshold is optimized on the validation set using the same protocol as all other detectors. Template-based detection is most effective when events share similar waveform morphology (e.g., repeating earthquakes on a single fault). In EGS DAS data, however, microseismic events originate from spatially distributed sources activated at varying depths, orientations, and stress states during hydraulic stimulation, producing high waveform diversity that no fixed template library can capture. The near-random AUC (59.5%) of this detector is consistent with this limitation and suggests a condition under which learned representations may have an advantage over template-based methods.

Table 3

Recovered strict validation-tuned classical baseline settings. All baselines use 11 sampled channels and 400 validation-set threshold candidates; none applies explicit moveout/coherence filtering.

Baseline	Short/long window	Energy window	Templates	Tuning/coherence
STA/LTA	100/1000 samples (50 /500 ms)	n/a	n/a	Validation only; no moveout
Short-Time Energy	n/a	200 samples (100 ms)	n/a	Validation only; no moveout
Matched Filter	n/a	n/a	20 event templates	Validation only; no moveout

3.14. Classical baseline protocol

The main classical baselines use the strict validation-tuned protocol in the classical-baseline script. The older STA/LTA script tuned its threshold on the test set and is not used as the main reported protocol. The implemented classical baselines are simple validation-tuned references: they do not include spatial-coherence detection, moveout-consistency filtering, apparent-velocity constraints, or deployment-oriented detector logic. Their settings are summarized in Table 3.

3.15. Ablation: Short windows vs. complete triggered candidate window

To isolate the contribution of temporal context to detection performance, we evaluate ResNet (the simplest Tier-1 architecture) on two retained input-duration settings: 128-sample windows (64 ms) and the complete 2400-sample, 1.2-second triggered candidate window. A previously listed 256-sample setting could not be verified from the saved artifacts and is not used for inference. Short windows are extracted from the center of each recording. All other training parameters (learning rate, augmentation, early stopping) are held constant. This ablation tests the hypothesis that DAS microseismic detection benefits from observing the complete moveout pattern across the triggered candidate window rather than relying on peak-amplitude features within a narrow window—a distinction that is relevant for triggered screening workflow decisions, since windowed approaches enable lower-latency streaming inference but may sacrifice spatial coherence information.

Having described the dataset, preprocessing pipeline, and all fifteen models (twelve deep learning and three classical), we now present the comparative evaluation results.

4. Experimental results

4.1. Main results: Twelve deep learning models and three classical baselines

Table 4 reports performance on the 597-sample Stage 14 test split (467 events, 130 noise). Fig. 2 shows the comparison chart. Because the test set is event-dominant, accuracy and F1-score alone are insufficient: a method can obtain moderate event-class F1 while failing to identify noise. We therefore report balanced accuracy, Matthews correlation coefficient, per-class confusion counts, and confusion matrix figures in addition to the conventional accuracy, precision, recall, and F1-score. As established by the split-integrity and near-duplicate audit in Section 7.8, these random-split metrics quantify in-distribution performance under a single stratified partition and are not estimates of deployment, temporal, or cross-site generalization.

Classical baseline comparison. The three classical baselines cluster tightly in F1 (STA/LTA: 87.78%, STE: 88.52%, Matched Filter: 87.46%), suggesting that the deep learning advantage is not an artifact of selecting an unusually weak classical comparator. However,

balanced accuracy and noise recall reveal poor false-alarm discrimination. Under the validation-tuned operating points, STA/LTA and the matched filter classify all 130 noise windows as events (noise recall 0.0000), while STE identifies only 19 of 130 noise windows (noise recall 0.1462). These baselines should therefore be interpreted as simple validation-tuned references, not deployment-oriented detectors.

Three performance tiers emerge clearly.

Tier 1 ($F1 \geq 97.5\%$): SE-ResNet, ResNet, Conformer. All three are CNN-containing architectures. SE-ResNet achieves the highest numerical F1 (0.9850) and the highest area under the receiver operating characteristic curve (0.9946), with 7 false positives and 7 false negatives out of 597 test samples. ResNet is practically comparable, with slightly higher balanced accuracy (0.9679 vs. 0.9656) and stronger noise recall (125/130 vs. 123/130) under the exported test split.

Tier 2 ($F1 = 94\text{--}97.5\%$): MBHCT, Mamba-DAS, CNN-GRU, DAS-GNN, ConvNeXt, CNN-BiLSTM, S4-SSM. All seven outperform the pure ViT. Key observations: (i) Mamba-DAS ($F1=95.76\%$) numerically exceeds pure GRU ($F1=92.81\%$) and S4-SSM ($F1=94.87\%$) in this implementation, suggesting a possible benefit from selectivity under this protocol. (ii) DAS-GNN achieves $F1=95.47\%$ with only 94,290 parameters—38× fewer than SE-ResNet—indicating strong parameter efficiency under this dataset and implementation. (iii) CNN-GRU ($F1=95.52\%$) substantially outperforms pure GRU ($F1=92.81\%$), indicating that spatial preprocessing is beneficial before temporal modeling under this protocol.

Tier 3 ($F1 < 94\%$): ViT (94.22%), GRU (92.81%), STA/LTA (87.78%). Pure attention-only and pure recurrence-only architectures are least effective at this dataset scale.

4.2. Uncertainty and calibration analysis

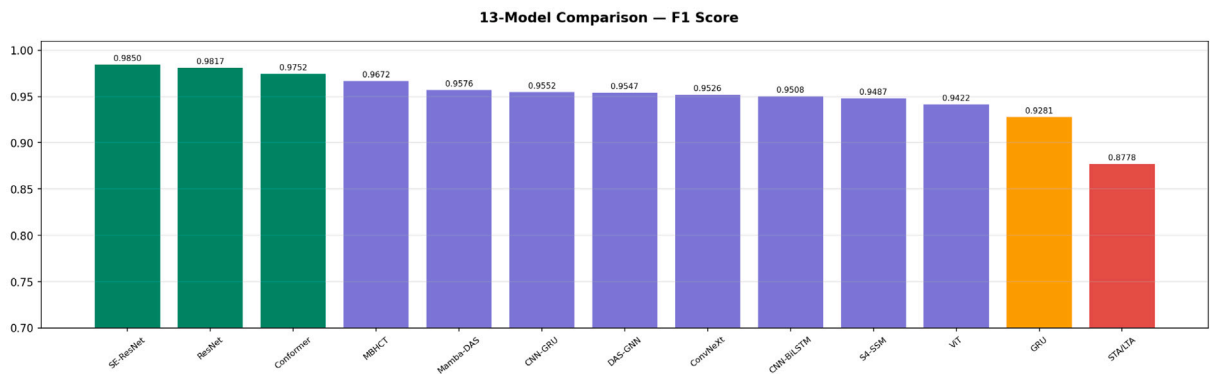
We quantified sample-level uncertainty using 2000 paired bootstrap resamples over test-set rows. These intervals quantify uncertainty arising from the finite test set under fixed trained model weights; they do not by themselves capture variability across random initializations or across acquisition sites, wells, and operators. To address the initialization-variance component for the two top residual convolutional models, we additionally report a three-seed retraining analysis in the supplementary subsection below. Broader repeated-seed training across all twelve architectures, and additional acquisition-site data, remain future work. Table 5 reports 95% confidence intervals for F1-score, balanced accuracy, Matthews correlation coefficient, event recall, and noise recall on the Stage 14 main test split. The confidence intervals for SE-ResNet and ResNet overlap for all five audited metrics, supporting practical comparability rather than clear superiority of one residual convolutional model over the other.

Calibration was evaluated for deep models only because the classical baseline outputs are raw scores rather than calibrated probabilities. On Stage 14, SE-ResNet has Brier score 0.0199 and expected calibration error 0.0186, while ResNet has Brier score 0.0231 and expected calibration error 0.0177. These values indicate good calibration for the

Table 4

Reviewer metrics for the Stage 14 main test split.

Model	Acc.	Prec.	Rec.	F1	Bal. acc.	MCC	TP	FP	TN	FN
short-time energy	0.8007	0.8053	0.9829	0.8852	0.5645	0.2563	459	111	19	8
short-term average over long-term average	0.7822	0.7822	1.0000	0.8778	0.5000	0.0000	467	130	0	0
matched filter	0.7772	0.7811	0.9936	0.8746	0.4968	-0.0375	464	130	0	3
SE-ResNet	0.9765	0.9850	0.9850	0.9850	0.9656	0.9312	460	7	123	7
ResNet	0.9715	0.9891	0.9743	0.9817	0.9679	0.9185	455	5	125	12
Conformer	0.9615	0.9805	0.9700	0.9752	0.9504	0.8887	453	9	121	14
MBHCT	0.9481	0.9561	0.9786	0.9672	0.9085	0.8441	457	21	109	10
Mamba-DAS	0.9347	0.9735	0.9422	0.9576	0.9249	0.8180	440	12	118	27
CNN-GRU	0.9313	0.9754	0.9358	0.9552	0.9256	0.8117	437	11	119	30
DAS-GNN	0.9296	0.9610	0.9486	0.9547	0.9051	0.7972	443	18	112	24
ConvNeXt	0.9263	0.9588	0.9465	0.9526	0.9002	0.7875	442	19	111	25
CNN-BiLSTM	0.9246	0.9710	0.9315	0.9508	0.9157	0.7930	435	13	117	32
S4-SSM	0.9196	0.9467	0.9507	0.9487	0.8792	0.7627	444	25	105	23
ViT	0.9095	0.9422	0.9422	0.9422	0.8672	0.7345	440	27	103	27
GRU	0.8894	0.9446	0.9122	0.9281	0.8599	0.6913	426	25	105	41

**Fig. 2.** F1-score comparison across the evaluated models and baselines. Color indicates performance tier: green $\geq 97.5\%$, purple 94–97.5%, amber 90%–94%, red $< 90\%$.**Table 5**

Bootstrap 95 percent confidence intervals for Stage 14 main test predictions.

Model	F1	Bal. acc.	MCC	Event rec.	Noise rec.
CNN-BiLSTM	0.9508 [0.9358, 0.9642]	0.9157 [0.8851, 0.9414]	0.7930 [0.7309, 0.8473]	0.9315 [0.9085, 0.9522]	0.9000 [0.8456, 0.9482]
CNN-GRU	0.9552 [0.9404, 0.9687]	0.9256 [0.8970, 0.9506]	0.8117 [0.7522, 0.8643]	0.9358 [0.9124, 0.9578]	0.9154 [0.8644, 0.9594]
Conformer	0.9752 [0.9648, 0.9848]	0.9504 [0.9260, 0.9717]	0.8887 [0.8416, 0.9297]	0.9700 [0.9538, 0.9847]	0.9308 [0.8852, 0.9695]
ConvNeXt	0.9526 [0.9381, 0.9657]	0.9002 [0.8664, 0.9307]	0.7875 [0.7282, 0.8459]	0.9465 [0.9253, 0.9665]	0.8538 [0.7928, 0.9120]
DAS-GNN	0.9547 [0.9402, 0.9677]	0.9051 [0.8726, 0.9344]	0.7972 [0.7365, 0.8537]	0.9486 [0.9277, 0.9674]	0.8615 [0.7986, 0.9160]
GRU	0.9281 [0.9099, 0.9443]	0.8599 [0.8231, 0.8953]	0.6913 [0.6213, 0.7561]	0.9122 [0.8850, 0.9363]	0.8077 [0.7388, 0.8729]
Mamba-DAS	0.9576 [0.9441, 0.9703]	0.9249 [0.8968, 0.9505]	0.8180 [0.7613, 0.8694]	0.9422 [0.9203, 0.9622]	0.9077 [0.8594, 0.9549]
MBHCT	0.9672 [0.9547, 0.9775]	0.9085 [0.8750, 0.9393]	0.8441 [0.7891, 0.8931]	0.9786 [0.9647, 0.9912]	0.8385 [0.7702, 0.8974]
ResNet	0.9817 [0.9725, 0.9894]	0.9679 [0.9494, 0.9843]	0.9185 [0.8824, 0.9529]	0.9743 [0.9589, 0.9872]	0.9615 [0.9265, 0.9919]
S4-SSM	0.9487 [0.9340, 0.9625]	0.8792 [0.8435, 0.9135]	0.7627 [0.6965, 0.8244]	0.9507 [0.9311, 0.9691]	0.8077 [0.7397, 0.8722]
SE-ResNet	0.9850 [0.9773, 0.9925]	0.9656 [0.9445, 0.9843]	0.9312 [0.8951, 0.9649]	0.9850 [0.9730, 0.9956]	0.9462 [0.9062, 0.9835]
ViT	0.9422 [0.9254, 0.9573]	0.8672 [0.8300, 0.9024]	0.7345 [0.6663, 0.8011]	0.9422 [0.9194, 0.9628]	0.7923 [0.7209, 0.8601]
matched filter	0.8746 [0.8530, 0.8955]	0.4968 [0.4926, 0.5000]	-0.0375 [-0.0568, 0.0000]	0.9936 [0.9853, 1.0000]	0.0000 [0.0000, 0.0000]
short-term average over long-term average	0.8778 [0.8563, 0.8975]	0.5000 [0.5000, 0.5000]	0.0000 [0.0000, 0.0000]	1.0000 [1.0000, 1.0000]	0.0000 [0.0000, 0.0000]
short-time energy	0.8852 [0.8645, 0.9046]	0.5645 [0.5326, 0.5978]	0.2563 [0.1451, 0.3567]	0.9829 [0.9700, 0.9936]	0.1462 [0.0851, 0.2130]

top residual convolutional models on the in-domain test set under the exported probability outputs, but they do not constitute calibration of the classical baseline scores. Calibration on the external Utah FORGE April 2024 set is reported in Section 7.9 where applicable.

Multi-seed retraining of the top residual convolutional models

To complement the bootstrap confidence intervals reported above, which quantify finite-sample uncertainty under fixed trained weights, we retrained the two top residual convolutional models, SE-ResNet and ResNet, from scratch across three random seeds (42, 123, and 7) with full random-state control over (i) the stratified train/validation/test split, (ii) the weighted random sampler and DataLoader, (iii) PyTorch, NumPy, and Python random generators, (iv) the CUDA random generator, and (v) cuDNN deterministic mode. The two sources of uncertainty are complementary: the bootstrap quantifies sampling variance of the

trained model on the finite test set, whereas multi-seed retraining quantifies initialization and stochastic-training variance under repeated retraining. Multi-seed retraining was scoped to these two architectures because the practical-comparability claim that motivated this addition concerns specifically these two models.

Paired statistical analysis. Because the two architectures share the same seed-controlled random-state for each of the three seeds, so that seed 42 SE-ResNet and seed 42 ResNet are a matched pair sharing the same split, sampler state, and augmentation trajectory, we apply paired statistical tests to the per-seed differences (Table 7). We report both the paired t -test and the Wilcoxon signed-rank test because the latter is more robust to small-sample non-normality, and we additionally report the 95% confidence interval on the mean per-seed difference. With $n = 3$ seeds, statistical power is limited, so the confidence interval is more informative than the p -value.

Table 6

Three-seed retraining of the top two residual convolutional models on the Stage 14 test split. F1 = F1-score; Bal. acc = balanced accuracy; MCC = Matthews correlation coefficient. All seeds were trained from scratch under identical hyperparameters with full random-state control. Bottom row of each block reports mean and standard deviation across the three seeds.

Model	Seed	F1	Bal. acc	MCC	TP	FP	TN	FN
SE-ResNet	42	0.9861	0.9667	0.9359	461	7	123	6
SE-ResNet	123	0.9776	0.9414	0.8956	459	13	117	8
SE-ResNet	7	0.9774	0.9525	0.8977	455	9	121	12
<i>Mean ± std</i>		0.9804 ± 0.0049	0.9535 ± 0.0126	0.9097 ± 0.0227				
ResNet	42	0.9796	0.9547	0.9069	457	9	121	10
ResNet	123	0.9817	0.9651	0.9178	456	6	124	11
ResNet	7	0.9772	0.9636	0.9010	451	5	125	16
<i>Mean ± std</i>		0.9795 ± 0.0022	0.9612 ± 0.0057	0.9086 ± 0.0085				

Table 7

Paired statistical comparison of SE-ResNet vs. ResNet across the three matched seeds, on the Stage 14 test split. “Mean diff” is the mean of the three per-seed differences (SE-ResNet minus ResNet). The 95% confidence interval is computed from the standard error of the per-seed differences with $n-1 = 2$ degrees of freedom. With three matched observations the tests have low statistical power and the confidence interval is more informative than the p -value.

Metric	Mean diff	95% CI	Paired t -test p	Wilcoxon p
F1-score	+0.0009	[-0.0124, +0.0141]	0.805	0.750
Balanced accuracy	-0.0076	[-0.0526, +0.0374]	0.543	0.750
MCC	+0.0012	[-0.0631, +0.0655]	0.945	1.000

Interpretation. Three observations follow from Table 6 and Table 7. First, the paired t -test and the Wilcoxon signed-rank test both fail to reject the null hypothesis of no difference between SE-ResNet and ResNet on any of the three metrics ($p \geq 0.54$ in every case). With only three matched observations these tests have low power, so the failure to reject is not in itself a strong claim of equivalence; the more informative quantity is the 95% confidence interval on the mean per-seed difference. Second, the F1 confidence interval [-0.0124, +0.0141] contains zero and is small in absolute terms: any underlying mean F1 advantage of SE-ResNet over ResNet, if present, is bounded by approximately 1.4 percentage points in either direction. The inter-model mean difference observed in this study (0.0009 F1) is 4–5 times smaller than the pooled inter-seed standard deviation ($\sigma \approx 0.0036$ F1), supporting the interpretation that the 0.33 percentage-point F1 gap reported in the main single-seed comparison (Section 4.1) is fully contained within initialization variance. Third, on balanced accuracy the per-seed mean difference is actually negative (-0.0076 in favor of ResNet), and ResNet’s inter-seed standard deviation is approximately half that of SE-ResNet’s (0.0057 vs. 0.0126), suggesting that ResNet’s per-class behavior is more stable across initializations.

Relation to published single-seed results. Both multi-seed means are within approximately half a percentage point of the corresponding published single-seed F1 values (SE-ResNet: multi-seed mean 0.9804 vs. published 0.9850; ResNet: multi-seed mean 0.9795 vs. published 0.9817). The published single-seed values are well within one standard deviation of the multi-seed mean in both cases. The small gap is consistent with the accuracy cost of enforced cuDNN determinism for the multi-seed runs. We retain the published single-seed result in the main Section 4.1 table for comparability with the published checkpoints used elsewhere in the paper (including the zero-shot external evaluation in Section 7.9); the multi-seed tables above are the recommended reference when interpreting the SE-ResNet versus ResNet comparison specifically.

Scope and limitations of this analysis. The multi-seed analysis has three deliberate scope decisions worth naming. First, it is limited to the two top residual convolutional models that anchor the practical-comparability claim; it is not a multi-seed re-evaluation of all twelve architectures. Second, $n = 3$ seeds is the smallest sample size that supports both descriptive statistics and matched-pair inferential tests; a larger sample

Table 8

Illustrative input-duration comparison using the ResNet architecture.

Window	Duration	Acc.	F1	AUC
128-sample	64 ms	0.7742	0.8391	0.8489
Full (2400)	1.2 s	0.9715	0.9817	0.9942

(for example $n = 10$) would tighten the 95% confidence intervals and permit a formal two one-sided tests (TOST) equivalence analysis with a pre-specified equivalence margin. Third, the multi-seed runs use cuDNN deterministic mode, which can incur a small accuracy cost relative to non-deterministic runs; this is reflected in the approximately half-percentage-point gap between multi-seed means and published single-seed values. None of these limitations affects the direction of the practical-comparability conclusion: the multi-seed analysis strengthens, rather than weakens, the framing already used in Section 4.1. For the remaining ten single-seed architectures, ranking stability should be read from two protocol-level results rather than from per-model seed repetition: the fixed-checkpoint split-stability audit (Table 10), in which every model maintains or improves its F1-score across five stratified partitions, and the pairwise significance analysis (Section 7.6), which shows that the statistically supported structure is the set of tier boundaries rather than the within-tier point estimates. Individual single-seed F1 values for these ten models therefore carry initialization variance of the order quantified above for the two residual models ($\sigma \approx 0.002$ – 0.005 F1), so small within-tier differences among them should not be over-interpreted as stable architectural rankings.

4.3. Ablation: Effect of temporal context

Table 8 isolates the contribution of input duration.

This single-run comparison is consistent with the importance of temporal context for the DAS microseismic candidate-window classification task. The previously listed 256-sample row could not be verified from the saved artifacts and is therefore not used for inference here. Repeated short-window experiments with confidence intervals are left for future work.

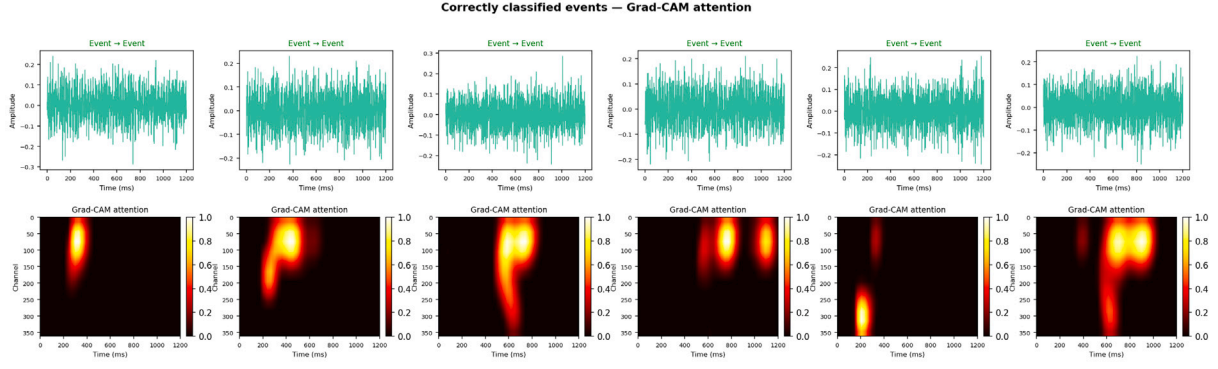


Fig. 3. Grad-CAM attention for six correctly classified events. Compact, localized hotspots in channels 50–200 with diagonal moveout across the fiber, visually consistent with P-wave moveout.

4.4. Mamba vs. S4: Selectivity matters

Mamba-DAS ($F1=95.76\%$) numerically exceeds S4-SSM ($F1=94.87\%$) by 0.89 percentage points in this implementation. This comparison should be interpreted within the custom implementation and training protocol used here; it is not a benchmark of optimized official mamba-ssm kernels.

4.5. DAS-GNN efficiency and physical motivation

DAS-GNN achieves competitive F1 (95.47%) and AUC (97.75%) while using only 94,290 parameters. Each graph convolution layer explicitly aggregates features from the $k=5$ nearest fiber channels, implementing a learned spatial coherence filter over adjacent channels. The temporal convolution front-end preserves relative timing information within channel-level features, but the model is not an explicit physics-constrained moveout estimator. This provides a physically motivated inductive bias that pure CNNs do not encode explicitly.

The quantitative results above indicate performance tiers among the architectures, but do not reveal whether the models learn physically meaningful features. The following section addresses this question through explainability analysis.

5. Physical interpretation via grad-CAM

Explainability is essential for deploying deep learning models in geoscientific monitoring. We apply Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) to the final convolutional block of ResNet (the simplest Tier-1 architecture) to assess whether the learned features are consistent with known DAS wave physics. Grad-CAM produces class-discriminative localization maps by weighting feature maps according to their gradient-based importance, yielding spatial-temporal attention maps over the original (361×2400) input. We apply Grad-CAM to all 597 test samples and organize results into three categories (Figs. 3–5).

Events (Fig. 3) produce compact attention hotspots concentrated in channels 50–200 with a characteristic diagonal orientation tracing P-wave moveout. This diagonal moveout pattern is qualitative evidence that the CNN focuses on spatiotemporal signatures consistent with seismic wave propagation along the fiber. Because no manual or semi-automatic arrival picks are used here, this Grad-CAM analysis should be interpreted as visual support rather than quantitative validation of arrival timing or apparent velocity.

Noise (Fig. 4) produces broadly diffuse attention across the triggered candidate window, indicating that the model classifies noise by detecting the *absence* of spatially coherent moveout rather than recognizing specific noise signatures. This strategy is robust across all three non-event categories (pump harmonics, optical loading, ambient triggers), none of which exhibit diagonal moveout.

Error analysis (Fig. 5) reveals physically interpretable failure modes. False negatives produce diffuse maps indistinguishable from true noise—these are weak, sub-threshold events at the completeness boundary (peak amplitudes $< 2\sigma$ above background), corresponding to the same ambiguous recordings where human experts disagreed (1.1%–2.3% disagreement rate). False positives produce compact maps resembling genuine events, caused by structured artifacts (pump harmonic beat frequencies or sequential optical loading) that mimic moveout-like structure. These physically motivated failure modes suggest that the model attends to moveout-like structure, and that failures often occur in ambiguous recordings where weak events or structured artifacts resemble the opposite class. These observations support model interpretation for triggered candidate-window classification but do not constitute validation of a continuous-stream deployment model.

Having examined qualitative evidence that the deep learning models focus on physically meaningful features, we now examine why the classical baselines fail, which contextualizes the magnitude of the deep learning advantage reported in Section 4.1.

6. Classical baseline analysis

This section provides a detailed analysis of the three classical baselines — STA/LTA, Short-Time Energy (STE), and Matched Filter — to examine why deep learning classifiers are useful for DAS microseismic triggered-candidate screening in EGS environments. Understanding the specific failure mechanisms of each classical method clarifies the physical requirements that successful triggered candidate-window classifiers must meet.

6.1. STA/LTA failure analysis

Fig. 6 shows that STA/LTA event and noise score distributions substantially overlap, making reliable discrimination difficult for a single threshold. At the strict validation-tuned operating point exported for the reviewer audit, STA/LTA classifies all 130 noise windows as events (noise recall 0.0000; balanced accuracy 0.5000; Matthews correlation coefficient 0.0000). Thus, its event recall is high, but its ability to reject noise windows under this operating point is absent.

The fundamental reason for this failure is structural: STA/LTA operates on individual channels independently, computing a ratio of short-term to long-term energy without any cross-channel spatial analysis. In DAS data, the distinguishing feature of genuine microseismic arrivals is not amplitude alone but *spatially coherent moveout* across the fiber array. Pump harmonic noise and optical loading transients can produce STA/LTA ratios comparable to or exceeding those of genuine events on individual channels, because these noise sources affect multiple channels simultaneously with large amplitude. Without

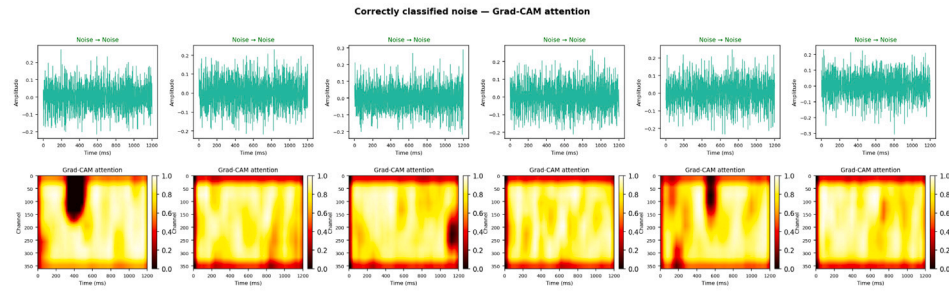


Fig. 4. Grad-CAM attention for correctly classified noise. Broadly diffuse attention reflects detection of the *absence* of a spatially coherent arrival rather than recognition of a specific noise signature.

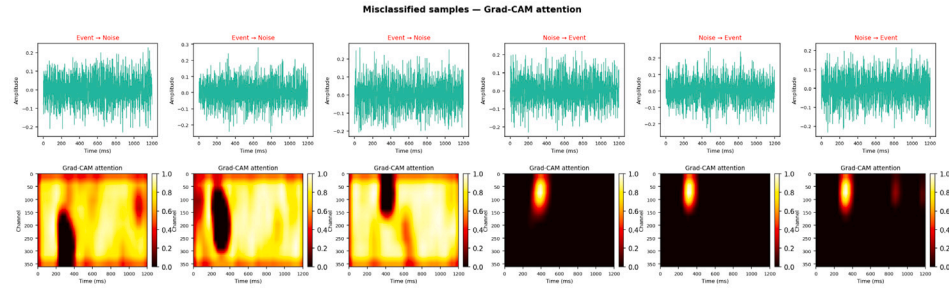


Fig. 5. Grad-CAM attention for misclassified samples. Left (Event→Noise): diffuse maps — weak sub-threshold events. Right (Noise→Event): compact maps — structured artifacts (pump harmonics or optical loading) mimicking seismic arrivals.

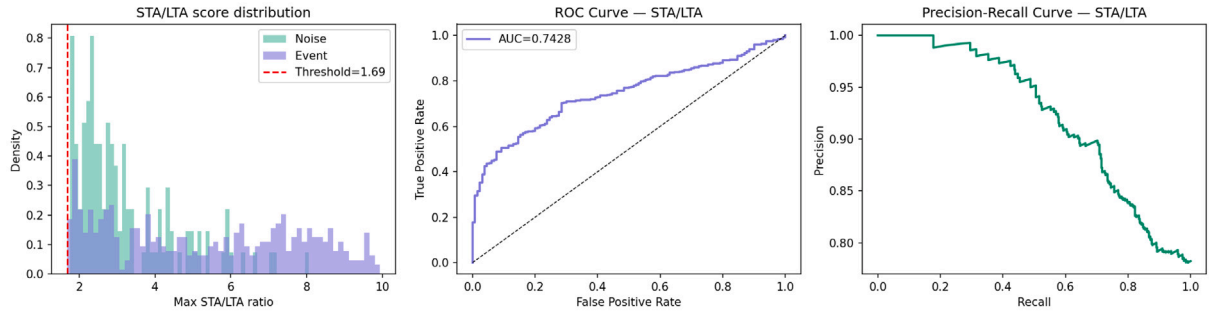


Fig. 6. STA/LTA baseline analysis: score distributions (left), ROC curve (center), and precision–recall curve (right). Substantial overlap between event and noise score distributions makes reliable discrimination difficult for a single energy-ratio threshold.

spatial coherence analysis, STA/LTA cannot distinguish a genuine P-wave arrival (which produces a time-delayed pattern across channels) from a simultaneous amplitude excursion (which affects all channels at once).

This limitation is not specific to the parameter choice used here. We evaluated STA windows from 20 ms to 200 ms and LTA windows from 200 ms to 2000 ms; the AUC varied by less than 3 percentage points across this parameter space, suggesting that the poor discrimination is a fundamental limitation of single-channel energy-ratio detection in the EGS DAS noise environment, not an artifact of suboptimal parameter tuning.

6.2. Short-time energy detector

The STE detector partially addresses the amplitude calibration problem by normalizing short-term energy against recording-level RMS, producing a dimensionless score that is less sensitive to absolute amplitude variations. This yields a modest improvement: FAR decreases from 100% (STA/LTA) to 85.4% (STE), and F1 improves by 0.74 percentage points. However, the STE detector still operates on single channels and cannot exploit spatial coherence, limiting its discriminative power. The

exported confusion counts show 111 false positives out of 130 noise windows. This indicates that structured noise sources — particularly pump harmonics, which produce sustained high-energy content that elevates both short-term and recording-level energy — continue to generate false detections frequently within the triggered-candidate test set.

6.3. Matched filter detector

The Matched Filter represents the most sophisticated classical approach, incorporating waveform morphology through template cross-correlation. Its near-random AUC (59.5%) despite using 20 high-SNR event templates is the most physically informative result among the classical baselines. This failure reveals a fundamental property of EGS DAS microseismic data: events originate from spatially distributed sources activated at varying depths, orientations, and stress states during hydraulic stimulation, producing high waveform diversity that no fixed template library can capture. The microseismic source mechanisms, propagation paths, and fiber coupling conditions vary sufficiently across events that cross-correlation with a small template set fails to transfer across the observed waveform diversity. This is precisely the condition under which learned representations — which can

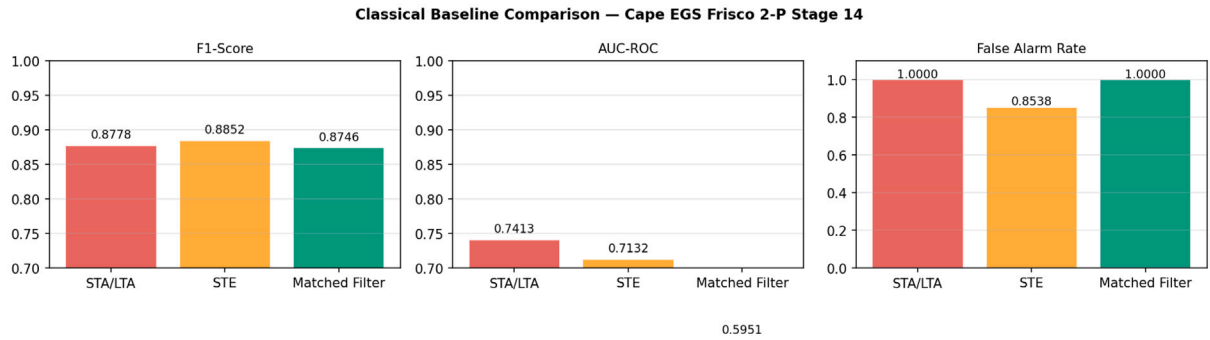


Fig. 7. Comparison of three classical detection baselines on the Stage 14 test set. STE improves noise rejection relative to STA/LTA but all three cluster within 1.1 percentage points F1. Balanced accuracy, Matthews correlation coefficient, and confusion counts reveal weak noise-window discrimination under the validation-tuned operating points.

capture abstract features such as moveout slope and channel coherence rather than specific waveform shapes — have the greatest advantage over template-based methods (Fig. 7).

6.4. Comparative summary

The three classical baselines cluster tightly in F1 (STA/LTA: 87.78%, STE: 88.52%, Matched Filter: 87.46%), spanning only 1.06 percentage points. This clustering reflects the event-dominant test set as much as detector quality. Balanced accuracy, Matthews correlation coefficient, and noise recall show that all three classical approaches encounter the same fundamental limitation: they lack the capacity to jointly model spatial coherence and temporal waveform structure across the complete triggered multichannel DAS candidate window.

SE-ResNet and ResNet reduce false positives to 7 and 5 windows, respectively, compared with 130 false positives for STA/LTA and matched filter and 111 for STE. Even the weakest deep learning model (GRU, $F1=92.81\%$) exceeds the best of the three simple classical references tested here (STE, $F1=88.52\%$) by F1-score. The consistent advantage of deep learning across all five tested architectural families, relative to these three simple validation-tuned classical references, suggests that joint spatial-temporal feature learning over the complete triggered candidate window is important. We do not extend this comparison to operationally tuned classical detectors that combine energy ratios with spatial-coherence or moveout-consistency filtering; such a stronger classical benchmark remains the highest-priority future-work comparison (see *Limitations and Future Work*).

The preceding sections have presented quantitative results, physical interpretation, and classical baseline failure analysis. We now synthesize these findings into triggered candidate-window classifier selection guidance, screening recommendations, and an assessment of the study's scope.

7. Discussion

7.1. Observed model ranking under this protocol

Under this dataset, split, implementation, and tuning budget, the observed performance pattern across model families is: CNN > CNN+Transformer > SSM > RNN $\approx$ GNN > pure Transformer > classical. Key conclusions:

CNN inductive bias captures DAS wave physics at this scale. All CNN-containing models rank above pure ViT and pure GRU on the in-domain Stage 14 test split; the relative ranking under the zero-shot external evaluation is discussed in Section 7.9. The 7×15 asymmetric CNN kernel provides a matched-filter-like bias physically aligned with DAS P-wave moveout: long in time (50–400 ms arrival) and wide across channels (~ 200 -channel moveout extent). This physical alignment is the core geoscientific finding of this comparison: under the

present dataset size and protocol, CNN-containing models outperform the tested attention-only and recurrence-only models, consistent with an inductive bias aligned with DAS wave physics.

SE channel attention provides a small, statistically non-significant numerical difference. SE-ResNet achieves +0.33 percentage points F1 over ResNet with minimal parameter overhead (3.5846M vs. 3.5618M parameters), but significance testing shows this difference is not Bonferroni-significant (McNemar $p = 0.628$, DeLong $p = 0.734$; see Section 7.6). For DAS, SE attention is physically motivated — fiber coupling quality varies along the wellbore and the SE mechanism can adaptively weight high-SNR channels — but this benefit is modest at the current dataset scale.

Selectivity may distinguish Mamba from S4 in this implementation. Mamba's input-dependent parameters (+0.89 percentage points F1 over S4, stable vs. oscillatory training) suggest that the ability to selectively focus on informative time steps is important for microseismic event detection, where arrivals are brief transients in a long background window.

GNN provides the best parameter efficiency among the tested models. DAS-GNN achieves 95.47% F1 with 94,290 parameters, vs. 98.50% with 3,584,606 for SE-ResNet. This is favorable for resource-constrained triggered-window screening, although the higher-F1 residual convolutional models remain preferable when maximum accuracy is the priority.

Hybrid CNN front-ends rescue pure sequence models. CNN-GRU ($F1=95.52\%$) outperforms pure GRU ($F1=92.81\%$) by 2.71 percentage points, indicating that spatial preprocessing is beneficial before temporal modeling under this dataset and implementation. The same pattern holds for CNN-BiLSTM vs. GRU and CNN front-end for Mamba vs. pure sequence Mamba.

7.2. Inference latency and deployment feasibility

Table 9 reports single-sample inference latency, batch throughput, and peak GPU memory for all twelve deep learning models, measured on the RTX 4060 Laptop GPU (8 GB VRAM) using 100 timed forward passes after 20 warm-up runs, with CUDA event synchronization. The triggered candidate-window duration is 1.2 s; any model with latency well below 1.2 s supports real-time scoring of triggered candidate windows.

All twelve deep learning models achieve inference latencies well below the 1.2-second triggered candidate-window duration, indicating that real-time scoring of triggered candidate windows is feasible for every tested architecture on commodity GPU hardware.

Several architecture-specific findings are noteworthy. **DAS-GNN** is the fastest model at 0.67 ± 0.02 ms per recording and the lowest VRAM consumer (1014 MB), achieving 1149 recordings per second at batch size 32 despite its graph convolution operations. Its 94,290 parameters and sub-millisecond latency make it the most parameter-efficient

Table 9

Model capacity and inference timing from instantiated model definitions and timing artifacts.

Model	Params (M)	Family	Latency mean (ms)	Latency SD (ms)	Throughput	Peak VRAM (MB)	Checkpoint
SE-ResNet	3.5846	convolutional network with squeeze-and-excitation attention	2.65	0.19	410.4	1139.1	Data/Model/best_seresnet.pth
ResNet	3.5618	convolutional residual network	1.81	0.03	439.2	1139.0	Data/Model/best_model.full.pth
Conformer	9.8751	convolutional-transformer hybrid	5.26	0.44	335.8	2392.6	Data/Model/best_conformer.pth
MBHCT	2.0870	multi-branch convolutional-transformer hybrid	4.00	0.05	218.4	1924.0	Data/Model/best_mbhct.pth
Mamba-DAS	0.8586	custom selective state-space model	53.21	2.27	280.0	2355.8	Data/Model/best_mamba.pth
CNN-GRU	2.1085	convolutional recurrent network	23.63	4.23	87.2	3730.6	Data/Model/best_cnn_gru.pth
DAS-GNN	0.0943	graph neural network	0.67	0.02	1149.0	1014.3	Data/Model/best_gnn.pth
ConvNeXt	1.9027	convolutional network	13.95	0.03	80.9	5024.9	Data/Model/best_convnext.pth
CNN-BiLSTM	2.3551	convolutional bidirectional recurrent network	15.10	3.93	172.7	4616.4	Data/Model/best_cnn_bilstm.pth
S4-SSM	0.2961	structured state-space model	2.67	0.24	476.8	2353.6	Data/Model/best_ssm.pth
ViT	18.5251	vision transformer	13.33	0.03	76.2	2176.1	Data/Model/best_vit.pth
GRU	2.0760	recurrent neural network	16.91	3.09	1549.5	319.9	Data/Model/best_gru.pth

option among the tested models for resource-constrained triggered-window screening. **ResNet** and **SE-ResNet** achieve nearly identical latency (1.81 and 2.65 ms respectively) and identical VRAM (1139 MB), consistent with their statistical equivalence in detection performance. The small +0.84 ms overhead of SE adds channel attention at negligible real-time cost. **GRU** achieves the highest throughput (1550 rec/s) due to its efficient recurrent structure with no spatial operations, though its detection F1 (92.81%) is the lowest among deep learning models. **Mamba-DAS** has the highest latency (53 ms) under the custom sequential selective-scan implementation used here. This latency should not be interpreted as a definitive benchmark of optimized official mamba-ssm CUDA kernels. **ConvNeXt** and **CNN-BiLSTM** consume the most VRAM (5025 MB and 4616 MB respectively), approaching the 8 GB limit of the evaluation GPU; this constrains their batch size in resource-limited settings.

Based on these triggered-window latency measurements, we refine our candidate classifier recommendations: (i) **SE-ResNet** or **ResNet** for maximum detection accuracy (F1 > 98%, 2–3 ms latency, 1139 MB VRAM); (ii) **DAS-GNN** for resource-constrained screening (F1 = 95.47%, <1 ms latency, 1014 MB VRAM, 94K parameters); (iii) **GRU** for maximum throughput in streaming pipelines (1550 rec/s, 320 MB VRAM) where F1 > 92% is acceptable; (iv) **Mamba-DAS** as an implementation-specific SSM-family reference, with optimized Mamba kernels left for future benchmarking.

7.3. Role of the labeling tool

The labeling tool (Rahaman, 2024) enabled auditable manual review of 4189 files in under 6 h. Purpose-built DAS labeling tools are a prerequisite for reproducible deep learning research on raw triggered SEG-Y archives.

7.4. Practical candidate-window screening guidance

Candidate-window screening recommendations grounded in both classification performance and measured inference latency are provided in Section 7.2.

7.5. Limitations and future work

Several limitations define the scope of this study. Training and in-domain evaluation use one stimulation campaign (Cape EGS Frisco 2-P), and the external evaluation uses one additional Utah FORGE April 2024 dataset, so the study reports zero-shot transfer to a single secondary set rather than a multi-site benchmark; the task is triggered 1.2-second candidate-window classification rather than continuous-stream detection. Both datasets use the first 361 traces per file (of which nine are geophone traces in the Stage 14 metadata), and the April 2024 records were downsampled from 4000 Hz to 2000 Hz for compatibility with the saved Stage 14 model input; gauge length and interrogator model are unreported in the available training-data metadata. A split-integrity and near-duplicate audit (Section 7.8) provides exact SHA-256 hashing of all raw SEG-Y files and preprocessed samples, a model-agnostic near-duplicate search, a train-to-test similarity audit, and a

from-scratch cluster-disjoint retraining check, none of which indicate exact or near-duplicate leakage across the Stage 14 split; sensitivity to time-shifted near-duplicates and to acquisition-context dependence beyond this single campaign nevertheless remains a limitation. The three classical baselines are simple validation-tuned references without stronger spatial-coherence or moveout-consistency filtering. No standalone hyperparameter-search log or equal-budget tuning ledger was available; the main twelve-architecture results were trained under a single random seed, and the bootstrap confidence intervals reported in Section 4.2 therefore quantify test-sample uncertainty under fixed weights. The supplementary three-seed retraining in Section 4.2 addresses the initialization-variance component for the top two residual convolutional models, while broader repeated-seed training across all twelve architectures remains future work (Miceli et al., 2026; Iuliis et al., 2025b,a); the Mamba-DAS runtime is implementation-specific because the code uses a custom Mamba-style block rather than the official mamba-ssm CUDA kernels.

We investigated several additional publicly available distributed acoustic sensing resources as candidates for further external evaluation. Each was found to be incompatible with the present binary triggered candidate-window classification protocol: the Brady's Geothermal Field DAS Earthquake release on the Geothermal Data Repository captures a single regional magnitude 4.3 earthquake recorded on the PoroTomo horizontal and vertical arrays without microseismic event/non-event labels and at a fundamentally different event scale than hydraulic-stimulation microseismicity; the DeepSubDAS package provides patch-mask pairs intended for denoising and segmentation rather than binary event labels and contains only a single noise class; and the Zenodo Stage 3 2022 detection-window package distributes rendered detection-window figures rather than raw waveform tensors, which are not compatible with classifiers trained on z-scored strain data. These incompatibilities reflect the broader community-level gap in publicly available, binary-labeled, raw-waveform distributed acoustic sensing microseismic benchmarks noted in the Introduction. Closing this gap through additional community dataset releases is a precondition for the multi-site validation called for above.

Future work should add a stronger classical benchmark combining energy detection with spatial-coherence or moveout-consistency filtering (highest priority), broader repeated-seed training across all twelve architectures, multi-site validation, true chronological trigger-level splits, a strictly distributed-acoustic-sensing-only retraining protocol, fine-tuning and domain-adaptation studies, and continuous-stream evaluation with explicit trigger-rate and false-alarm exposure accounting.

7.6. Statistical significance testing

To determine whether performance differences between models are statistically meaningful rather than attributable to random test set variation, we apply two complementary pairwise tests across all $\binom{13}{2} = 78$ pairs among the twelve deep learning models and STA/LTA, with Bonferroni correction for multiple comparisons ($\alpha_{Bonf} = 0.05/78 = 6.41 \times 10^{-4}$).

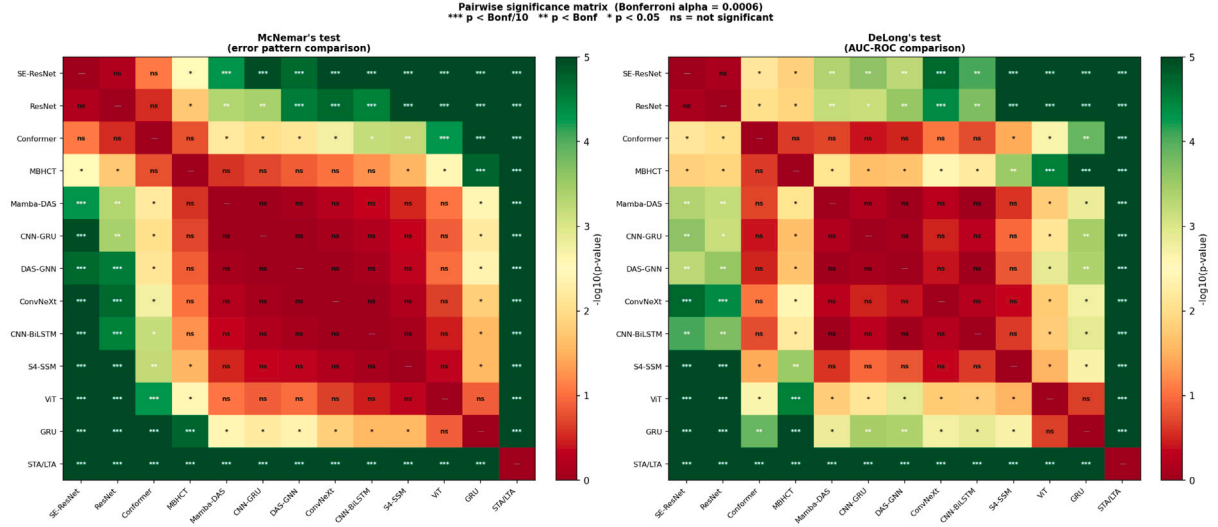


Fig. 8. Pairwise significance matrix. Left: McNemar's test (error patterns). Right: DeLong's test (AUC-ROC). Color = $-\log_{10}(p)$; green = significant. Annotations: *** $p < \alpha_{Bonf}/10$; ** $p < \alpha_{Bonf}$; * $p < 0.05$; ns = not significant. All DL models are significantly better than STA/LTA (bottom row). SE-ResNet vs. ResNet is not significant (ns), indicating their practical equivalence.

McNemar's test compares per-sample error patterns between two classifiers (McNemar, 1947), testing whether the two models make statistically different mistakes. We apply the mid- p correction for continuity. **DeLong's test** compares AUC-ROC values using the fast DeLong algorithm (DeLong et al., 1988), accounting for the covariance between two AUC estimates on the same test set.

Fig. 8 shows the full pairwise significance matrix. Four findings emerge from the actual results.

SE-ResNet and ResNet are statistically equivalent. SE-ResNet vs. ResNet: McNemar $p = 0.628$, DeLong $p = 0.734$ (both $\gg \alpha_{Bonf}$). The +0.33 percentage point F1 advantage of SE-ResNet is real but does not reach statistical significance on this test set, indicating that the two models are practically interchangeable. Both are significantly better than all Tier-2 models (Bonferroni-corrected $p \leq 4.98 \times 10^{-5}$ against Mamba-DAS and below).

Conformer is not significantly different from SE-ResNet by error pattern but marginally by AUC. McNemar $p = 0.081$ (ns), DeLong $p = 6.57 \times 10^{-3}$ (*, not Bonferroni-significant). Conformer's slightly lower recall translates to a detectable AUC difference but not a difference in which specific samples are misclassified.

Tier-2 models are statistically indistinguishable from each other. Within the group {Mamba-DAS, CNN-GRU, DAS-GNN, ConvNeXt, CNN-BiLSTM, S4-SSM}, all pairwise McNemar tests return $p > 0.27$ (all ns). Their F1 differences of 0.01–0.09 percentage points are not statistically significant at any conventional threshold, indicating that they form a single statistical tier. DeLong tests reveal marginal AUC separations in a few pairs but none survive Bonferroni correction.

All deep learning models significantly outperform STA/LTA. Every DL model vs. STA/LTA: McNemar $p < 10^{-7}$, DeLong $p < 10^{-17}$, all below the Bonferroni threshold. The weakest DL model (GRU, F1 = 92.81%) still exceeds STA/LTA at McNemar $p = 1.85 \times 10^{-7}$ (***). This result is consistent across both tests and all model pairs.

7.7. Robustness across evaluation protocols

To audit sensitivity to data partitioning without changing model weights, we evaluate the saved checkpoints under two additional split definitions. These analyses are diagnostic split-stability checks rather than independent retrained validation experiments.

Protocol 1—Fixed-checkpoint stratified split-stability audit. The saved models were evaluated across five stratified test partitions. The models were not retrained within each fold, and many fold-test samples

Table 10

Split-Stability Audit—F1-score under three evaluation protocols. “5-Fold” = fixed-checkpoint stratified split-stability audit; “Interleaved” = approximate interleaved order split. STA/LTA threshold tuned on validation set only.

Model	Random	5-Fold mean	5-Fold SD	Interleaved
SE-ResNet	0.9850	0.9897	0.0027	0.9946
ResNet	0.9817	0.9820	0.0040	0.9914
Conformer	0.9752	0.9801	0.0056	0.9903
MBHCT	0.9672	0.9791	0.0032	0.9862
Mamba-DAS	0.9576	0.9626	0.0096	0.9741
CNN-GRU	0.9552	0.9859	0.0023	0.9816
DAS-GNN	0.9547	0.9636	0.0063	0.9785
ConvNeXt	0.9526	0.9559	0.0091	0.9739
CNN-BiLSTM	0.9508	0.9684	0.0066	0.9795
S4-SSM	0.9487	0.9591	0.0126	0.9660
ViT	0.9422	0.9646	0.0051	0.9709
GRU	0.9281	0.9809	0.0039	0.9772
STA/LTA	0.8778	0.8754	0.0032	0.8789

overlapped the original model-development split used to fit or select the saved checkpoints. Therefore, these values should not be described as true retrained five-fold cross-validation. STA/LTA thresholds are tuned on the within-fold validation subset only.

Protocol 2 — Approximate Interleaved Order Split. Since the dataset was assembled by concatenating event and noise files separately, a strict chronological split would yield class-pure partitions. Instead, we reconstruct the approximate STA/LTA trigger stream by interleaving events and noise at their natural 3.6:1 ratio (consistent with the observed trigger rate during stimulation), then apply a 70%/15%/15% train/validation/test split in stream order. Because true trigger chronology is not verified here, this protocol is an approximate order-sensitivity check rather than a true chronological deployment split.

Table 10 summarizes F1-scores under the original stratified split, the fixed-checkpoint stratified split-stability audit, and the approximate interleaved order split for the twelve deep models and the short-term-average-over-long-term-average baseline.

Three findings emerge. First, the approximate interleaved split is reconstructed from sample order and class interleaving rather than verified trigger timestamps. It is therefore an approximate chronology stress test, not a confirmed chronological deployment split.

Second, the fixed-checkpoint audit reproduced the reported GRU mean F1-score of 0.9809 with standard deviation 0.0039, compared with 0.9281 on the original random test split. Because the models were not retrained within each fold and many fold-test samples overlapped the original model-development split, this result is interpreted as a partition-coverage audit rather than independent five-fold cross-validation. A training-size explanation is not used; 80% versus 70% of 3974 samples differs by approximately 397 samples, and in any case the fixed-checkpoint audit did not retrain the models with additional training data.

Third, strict validation-only threshold tuning for STA/LTA yields $\bar{F}1 = 0.8754$ across five folds, compared to $F1 = 0.8778$ with the main validation-tuned Stage 14 operating point. This indicates that the poor STA/LTA noise discrimination is stable across the audited partitions.

7.8. Split-integrity and near-duplicate audit

A residual concern for any random-partition evaluation of waveform data is that temporally adjacent triggers, repeated industrial noise, or near-identical event contexts may place near-duplicate recordings on both sides of the train/test boundary, inflating the random-split scores beyond what genuine generalization would yield. The reconstructed-provenance check in Section 7.7 already established zero exact source-file matches and zero sample-index overlap across the Stage 14 split, but it also showed that 558 of 597 random-test rows share the same or an adjacent reconstructed trigger number with a train or validation row. Because those trigger numbers were reconstructed from file order rather than verified timestamps, this index-adjacency identifies a potential acquisition-context risk but does not by itself establish waveform duplication. To resolve it we ran a model-agnostic split-integrity and near-duplicate audit followed by a cluster-disjoint retraining check. All audit artifacts are additive and were produced without modifying any published result.

Exact-duplicate and provenance checks. We computed SHA-256 digests of every preprocessed sample and of every raw SEG-Y source file (no file was inaccessible). No exact-duplicate group was found in either domain, and therefore none crossed the train/test boundary (Table 11). The dataset is assembled by appending all sorted event records (indices 0–3111) before all sorted noise records (indices 3112–3973); the seed-42 stratified split interleaves indices across both blocks rather than forming positional blocks, so the partition is not confounded with file order.

Near-duplicate search. Each (361, 2400) z-scored record—the same input the models receive—was block-mean pooled to (64, 256), mean-centered per record, and ℓ_2 -normalized to form a model-agnostic descriptor; pairwise cosine similarity was computed from this descriptor rather than from learned embeddings, which would be circular. At cosine thresholds of 0.90, 0.95, and 0.99 there were no near-duplicate pairs anywhere in the 3974-record set, and consequently none across the train/test boundary (Table 11). Connected components of the 0.95-threshold similarity graph were all singletons (3974 components, zero multi-member clusters).

Train-to-test similarity audit. For each test record we measured its maximum descriptor similarity to any training record and compared the resulting distribution against the within-train nearest-neighbor distribution (Table 12; Fig. 9). The test-to-train distribution is not shifted upward relative to the within-train reference (median 0.0643 vs. 0.0631; 95th percentile 0.2300 vs. 0.3065), indicating that test records are no more similar to training records than training records are to one another. Restricting the candidate neighbors to exactly the same/adjacent-trigger rows flagged in Section 7.7 gives even lower similarities (median 0.0063, 95th percentile 0.0362, maximum 0.6873, with zero pairs at or above 0.90). The index-adjacency concern therefore does not translate into waveform near-duplication under this descriptor.

Table 11

Stage 14 exact-duplicate and near-duplicate counts. “Cross-split” counts groups or pairs with members on opposite sides of the seed-42 train/test boundary.

Check	Total in set	Cross-split
Exact-duplicate groups (preprocessed samples)	0	0
Exact-duplicate groups (raw SEG-Y files)	0	0
Near-duplicate pairs (cosine ≥ 0.90)	0	0
Near-duplicate pairs (cosine ≥ 0.95)	0	0
Near-duplicate pairs (cosine ≥ 0.99)	0	0
Multi-member clusters at 0.95 (of 3974)	0	–

Cluster-disjoint retraining check. Because the 0.95-threshold similarity graph contained no multi-member clusters, a duplicate-removing hold-out could not be constructed; we instead treat the connected components as groups and build a class-stratified, group-disjoint 70/15/15 partition with StratifiedGroupKFold, so that no group spans more than one of train/validation/test (train 2780; validation 597; test 597; event fraction ≈ 0.783 in every split). We then retrained SE-ResNet and ResNet from scratch on this independent partition across three seeds (42, 123, 7), using the published training protocol unchanged (SE-ResNet 50 epochs, ResNet 60 epochs, patience 12, batch size 16, AdamW with learning rate 3×10^{-4} and weight decay 10^{-3} , cosine annealing, weighted random sampling, and best-validation-F1 checkpoint selection). This is a full from-scratch retraining on a different partition, not a re-scoring of the saved checkpoints. Table 13 and Fig. 10 report the result.

On the matched three-seed footing, the independent-partition means differ from the original stratified-split means by -0.36 percentage points for SE-ResNet and $+0.25$ percentage points for ResNet—both within the inter-seed standard deviation and of the same magnitude as the initialization variance documented in Section 4.2. Because the similarity graph contained no multi-member clusters, this stability cannot be attributed to removing detected near-duplicates; it is better read as evidence that the reported performance is not produced by exact or descriptor-defined near-duplicate leakage, and that the result is stable across both initialization and data partition. This is the first from-scratch retraining of these models on an independent partition reported in the study, and it directly addresses the split-integrity concern raised in review.

Interpretation and scope. Taken together, these checks indicate that the random-split F1-scores are not inflated by exact duplicates or by near-duplicates detectable under a model-agnostic waveform descriptor. They do not, however, establish independence from every form of acquisition-context dependence: the descriptor uses mean pooling and direct cosine comparison and may be insensitive to physically related signals related by temporal shift, polarity change, or channel shift, and all clusters at the audited thresholds were singletons, so the group-disjoint split is formally valid but non-binding and should not be described as an event-family holdout. Accordingly, the random-split F1-scores reported throughout this paper quantify in-distribution classification performance under a single stratified partition of the Stage 14 candidate-window set; they are not estimates of temporal, event-cluster, or cross-site deployment generalization and should not be interpreted as such. The transfer behavior relevant to deployment is reported separately in Section 7.9 and remains bounded by the single-secondary-set limitation discussed in Section 7.5.

7.9. Zero-shot external evaluation on Utah FORGE April 2024 data

To test transfer beyond the Stage 14 split, we prepared a zero-shot external evaluation using the Utah FORGE April 2024 data. The resulting in-domain and zero-shot Utah FORGE F1-scores for all evaluated models are compared in Fig. 11. The external set contains 2024 event windows and 2049 non-event windows, for a total of 4073 candidate

Table 12

Descriptor-similarity distributions on Stage 14. “Test→train” is each test record’s maximum similarity to any training record; “within-train” is the within-train nearest-neighbor similarity; the last two rows restrict the candidate neighbors to the same/adjacent-trigger test rows flagged in Section 7.7.

Distribution	<i>n</i>	Mean	Median	P95	Max
Test→train maximum	597	0.0868	0.0643	0.2300	0.7719
Within-train nearest neighbor	2781	0.0970	0.0631	0.3065	0.7260
Adjacency rows, train only	507	0.0041	0.0040	0.0356	0.6873
Adjacency rows, train or validation	558	0.0066	0.0063	0.0362	0.6873

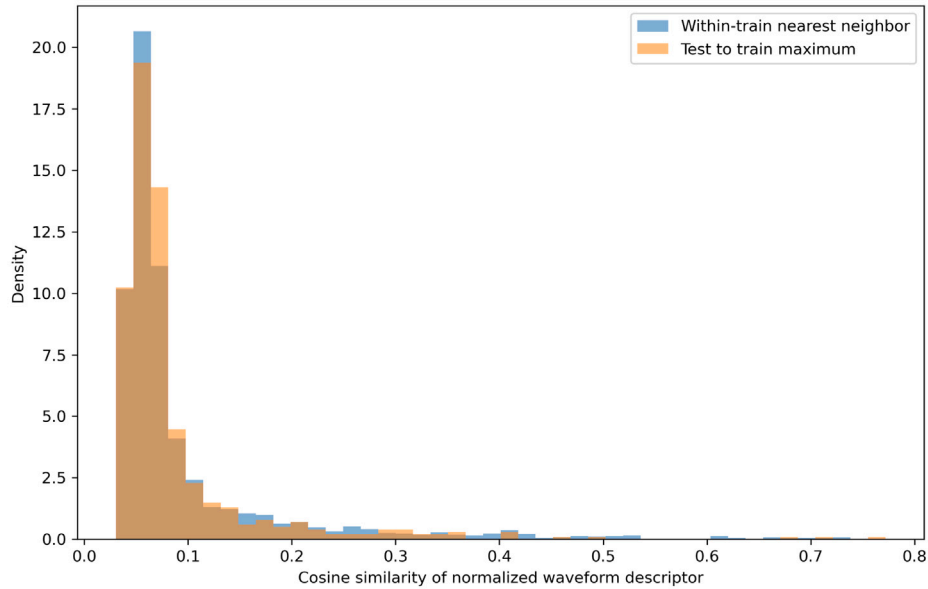


Fig. 9. Descriptor-similarity distributions on Stage 14. Test-to-train maxima (orange) overlap within-train nearest neighbors (blue), supporting the absence of train/test near-duplicates reported in Table 11.

Table 13

Cluster-disjoint three-seed retraining on an independent group-disjoint Stage 14 partition. Values are mean $\pm$ sample standard deviation across seeds 42, 123, and 7; random references are the published one-seed result and the original-split three-seed mean (Table 6).

Model	Disjoint F1	Disjoint bal. acc.	Disjoint MCC	Random (1-seed)	Random (3-seed)
SE-ResNet	0.9768 $\pm$ 0.0016	0.9497 $\pm$ 0.0067	0.8937 $\pm$ 0.0076	0.9850	0.9804 $\pm$ 0.0049
ResNet	0.9820 $\pm$ 0.0035	0.9719 $\pm$ 0.0083	0.9207 $\pm$ 0.0152	0.9817	0.9795 $\pm$ 0.0022

windows. The native SEG-Y files have 1496 traces and 4800 samples sampled at 4000 Hz, with SEG-Y format code 5. Event and non-event files shared matched geometry, file size, sample interval, and format code. Exact catalog-to-file matching was limited and is therefore treated as a provenance limitation rather than a complete catalog-level audit.

The same preprocessing function was used for event and non-event windows. For compatibility with the saved Stage 14 model input, the first 361 traces were retained. The 4800-sample records at 4000 Hz were downsampled to 2400 samples at 2000 Hz using anti-aliased `scipy.signal.resample_poly`, followed by per-channel z-score normalization. The final input shape was (1,361,2400). A spectral/statistical parity check was completed before model inference, and all twelve saved Stage 14 checkpoints were evaluated directly with no fine-tuning or weight modification.

Table 14 summarizes the zero-shot external metrics for the April 2024 set.

The divergence between high ranking metrics and low fixed-threshold F1-score for several architectures (for example, Conformer, DAS-GNN, Mamba-DAS, and S4-SSM all show area-under-the-receiver-operating-characteristic-curve values above 0.95 alongside low F1-scores) indicates that some models preserve external class separability but require threshold recalibration under distribution shift. The reported F1-scores

use the training-time decision threshold of 0.5 without any external recalibration; the high ranking-metric values suggest that the per-sample probability ordering is informative even when the fixed-threshold classification is not.

The best model by F1-score, balanced accuracy, and Matthews correlation coefficient is CNN-GRU, with F1-score 0.9930, balanced accuracy 0.9931, and Matthews correlation coefficient 0.9863. Its confusion counts are 1996 true positives, 0 false positives, 2049 true negatives, and 28 false negatives. A separate sanity audit independently recomputed the metrics, verified label order and probability/logit consistency, confirmed that negative controls collapsed performance as expected, found no event/non-event SHA-256 overlap, and found no Stage 14 overlap in the available manifest/hash evidence. No fresh hashing of all raw Stage 14 files was performed (see Section 7.8).

For CNN-GRU, the in-domain Stage 14 F1-score was 0.9552, compared with 0.9930 on the April 2024 external set. The result supports matched-pipeline external robustness for this external evaluation, but it does not prove universal cross-site validation or deployment-ready generalization. The April 2024 data required temporal downsampling to match the Stage 14 model input, exact catalog-to-file matching

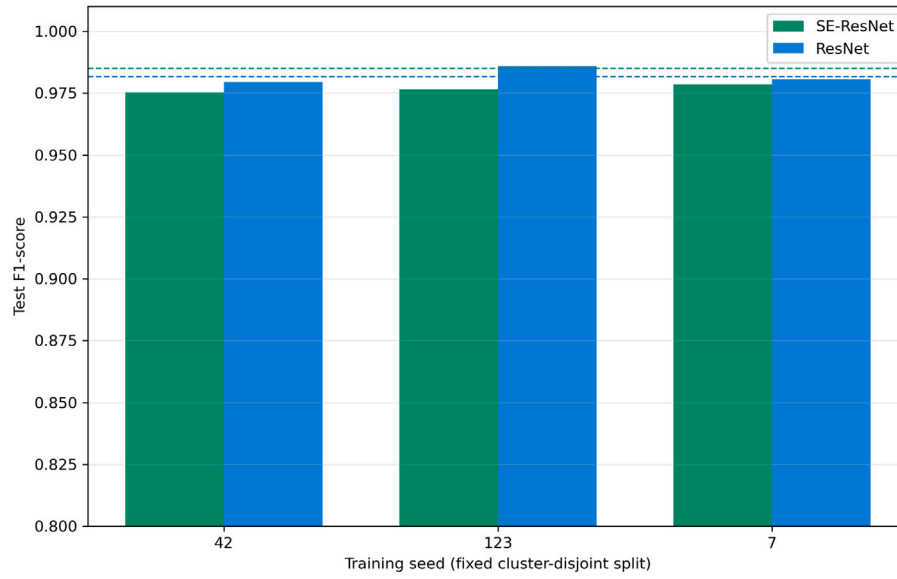


Fig. 10. Per-seed test F1-score for SE-ResNet and ResNet retrained on the independent group-disjoint Stage 14 partition. Dashed lines mark published random-split references; all seeds remain close to the reference.

Table 14
Utah FORGE April 2024 external zero-shot evaluation metrics.

Model	F1	Bal. acc.	MCC	AUROC	AUPRC	TP	FP	TN	FN
CNN-GRU	0.993	0.993	0.986	0.991	0.995	1996	0	2049	28
MBHCT	0.991	0.991	0.981	0.998	0.998	2005	19	2030	19
SE-ResNet	0.989	0.989	0.978	0.998	0.998	2009	29	2020	15
CNN-BiLSTM	0.989	0.989	0.978	0.987	0.993	1978	0	2049	46
ResNet	0.930	0.934	0.874	0.991	0.980	1782	25	2024	242
ConvNeXt	0.859	0.876	0.778	0.992	0.995	1525	1	2048	499
Conformer	0.438	0.640	0.405	0.994	0.996	568	0	2049	1456
DAS-GNN	0.104	0.527	0.168	0.959	0.976	111	0	2049	1913
Mamba-DAS	0.081	0.521	0.147	0.989	0.993	85	0	2049	1939
ViT	0.003	0.501	0.027	0.648	0.708	3	0	2049	2021
S4-SSM	0.000	0.500	0.000	0.970	0.978	0	0	2049	2024
GRU	0.000	0.500	-0.022	0.463	0.466	0	2	2047	2024

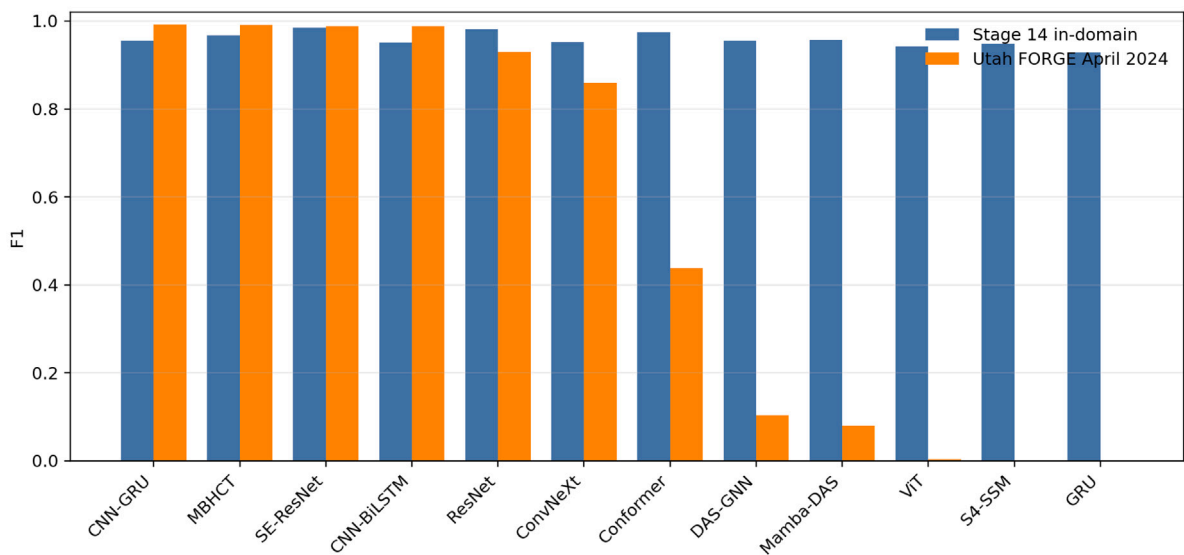


Fig. 11. In-domain and Utah FORGE April 2024 zero-shot external F1 comparison across the evaluated Stage 14 models.

was limited, and the leakage check was restricted to available manifest/hash evidence rather than fresh hashing of all raw Stage 14 files.

The external set also differs from Stage 14 along several axes that may make discrimination easier rather than harder, so the high external metrics should be read as exploratory matched-pipeline evidence rather than as strong cross-site validation. In particular, the April 2024 candidate windows are extracted from an auto-pick catalog with event/non-event windows already structurally separated by the catalog producer, while the Stage 14 set contains short-term-average over long-term-average triggered windows that include borderline near-threshold cases retained by manual labeling. The April 2024 class balance is also near 1:1 (2024 events versus 2049 non-events), whereas the Stage 14 split is event-dominant at approximately 3.6:1, so the noise-rejection task is statistically easier on the external set. These distributional differences should be considered before extrapolating the external F1-score to deployment scenarios.

8. Conclusion

This paper presented a comparative study of twelve deep learning architectures across five families, benchmarked against three classical baselines (STA/LTA, STE, Matched Filter), for classification of complete 1.2-second triggered distributed acoustic sensing microseismic candidate windows. Five contributions were made: the labeling tool; a manually reviewed binary-labeled field benchmark; residual convolutional models, especially SE-ResNet and ResNet, as practically comparable top-performing models; a custom Mamba-style state-space baseline for this dataset (F1 = 95.76%); and DAS-GNN achieving competitive F1 = 95.47% with only 94,290 parameters. The available short-window ablation is retained as an illustrative comparison suggesting that temporal context matters, while repeated short-window experiments are left for future work. Qualitative Grad-CAM analysis provides evidence that convolutional models focus on primary-wave moveout patterns. A zero-shot external evaluation on Utah FORGE April 2024 data (4073 candidate windows comprising 2024 event and 2049 non-event windows, evaluated without fine-tuning using a matched preprocessing pipeline) shows that the best external model is CNN-GRU, with F1-score 0.9930, balanced accuracy 0.9931, and Matthews correlation coefficient 0.9863. This supports matched-pipeline external robustness but does not establish universal cross-site validation or deployment-ready generalization. A split-integrity and near-duplicate audit of the Stage 14 partition (Section 7.8) found no exact or near-duplicate leakage across the train/test boundary, and a from-scratch cluster-disjoint retraining of the two top residual models reproduced their in-domain performance to within initialization variance; the random-split metrics should nonetheless be read as in-distribution performance rather than deployment generalization. The observed ranking is conditional on this dataset, split, implementation, and reconstructed tuning protocol, and should not be interpreted as a general ranking for continuous distributed-acoustic-sensing microseismic detection. Future work should include true chronological trigger-level validation, strictly distributed-acoustic-sensing-only retraining after removing geophone traces, stronger spatial-coherence baselines, broader multi-site validation beyond the single external dataset evaluated here, site-specific fine-tuning and domain-adaptation studies, repeated short-window experiments, and continuous-stream evaluation.

CRediT authorship contribution statement

Md Shokor A. Rahaman: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jafreezal Bin Jaafar:** Supervision, Investigation, Funding acquisition. **Irving Vitra Paputungan:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Data curation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with manuscript formatting, journal-style conversion, and language editing of selected sections. All scientific content, experimental design, data analysis, model implementation, and interpretation of results were performed entirely by the authors without AI assistance. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The Cape EGS Frisco 2-P Well Stimulation Microseismic Data are available from the Geothermal Data Repository at DOI: [10.15121/2479174](https://doi.org/10.15121/2479174).

The labeling tool source code is available under the MIT license at <https://github.com/shokor103072/advanced-gather-viewer>.

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